Knot Invariant: Classification, Categorification And Generalization

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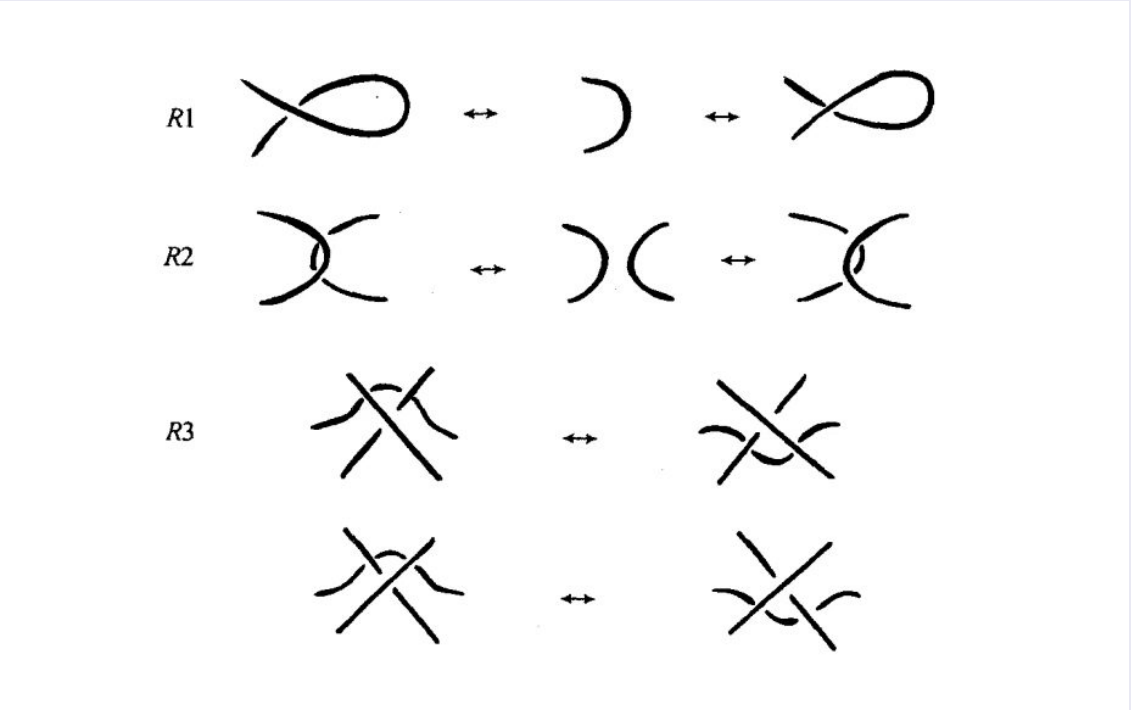
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Knots And Links

What is knots and links?

A knot is a closed curve with its two ends joined together and no self-intersections. If one knot can be continuously deformed into another, the two knots are said to be **ambient isotopic**.

Continuously deformed—Reidemeister move



Knots And Links

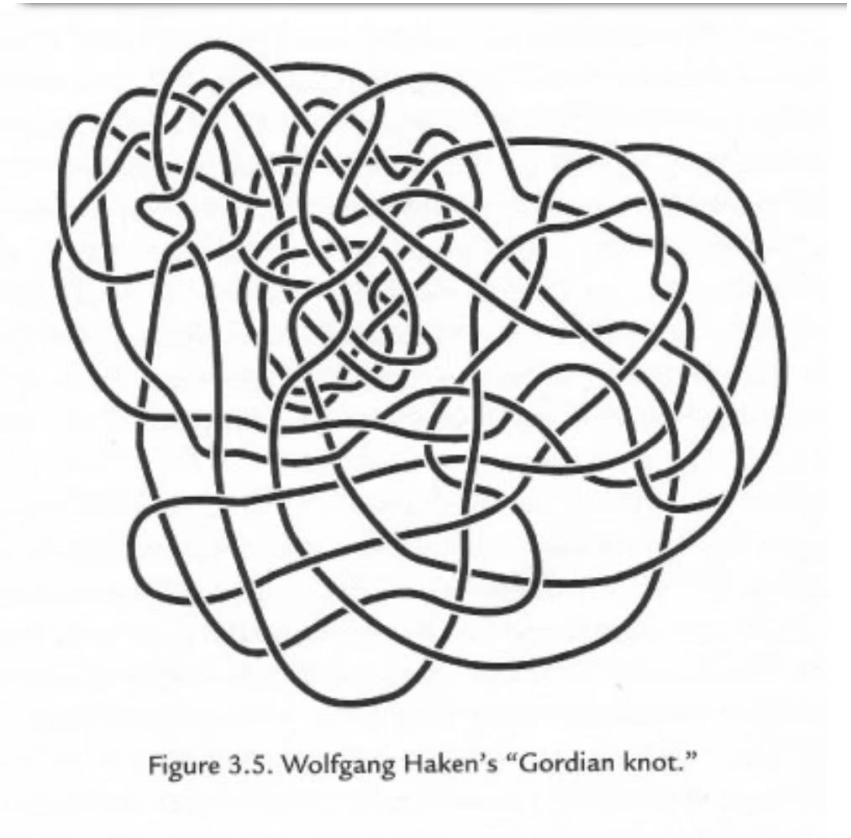


Figure 3.5. Wolfgang Haken's "Gordian knot."

图 1: Trivial Knot

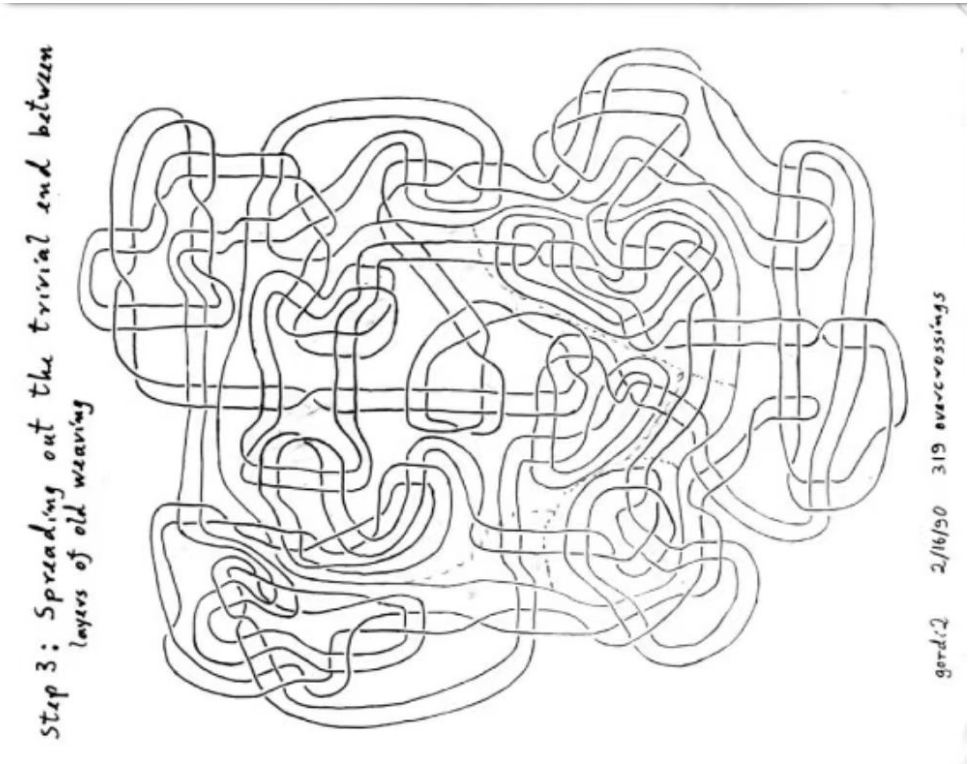


图 2: Trivial Knot

Knots And Links

Some Perspectives on Knot Invariants

(1)**Analysis**: It is a common approach to find invariants or certain important structures via solutions to nonlinear PDEs. Recent Paper: (Quantum equations for knots) https://arxiv.org/abs/2509.08423.
(2)**Algebra**: Polynomial invariants (Jones polynomial), homology invariants (Khovanov homology), knot quandle, etc. We will introduce it later.
(3)**Topology**: Seifert Genus, hyperbolic volume, etc.
(4)**Physics**: Charged Knots and Mobius Energy (Picture 3) , Some TQFT-related invariants, etc.

Two approaches to understanding knots

A. **Intrinsic viewpoint:** Untie or reconstruct the knot itself.
B. **Extrinsic viewpoint:** Properties of knots in the 3-sphere and its complement.

My thoughts on knots

(1) Unknotting is an art. (By the skein relation)
(2) Everything can be categorized. (By the link homologies)
(3) Surgery coexists with obstruction.

0_1	3_1	4_1	5_1	5_2	6_1	6_2	6_3	7_1	7_2	7_3	8_{19}
0	70.4	104.9	126.8	134.6	162.8	168.5	172.9	181.0	190.3	192.7	197.0
7_4	7_5	7_6	8_{20}	7_7	8_{21}	8_1	8_3	8_2	8_4	8_6	8_7
197.7	199.7	203.7	203.9	207.1	209.5	217.4	220.9	224.1	226.2	228.6	231.3
8_8	8_9	8_{11}	8_5	9_1	8_{12}	8_{13}	8_{10}	8_{14}	8_{15}	8_{16}	8_{17}
232.4	233.2	233.7	234.0	234.6	235.1	237.9	238.7	238.7	243.1	264.5	265.1

图 3: Mobius Energy (These knots are shown in the next figure)

Knots And Links

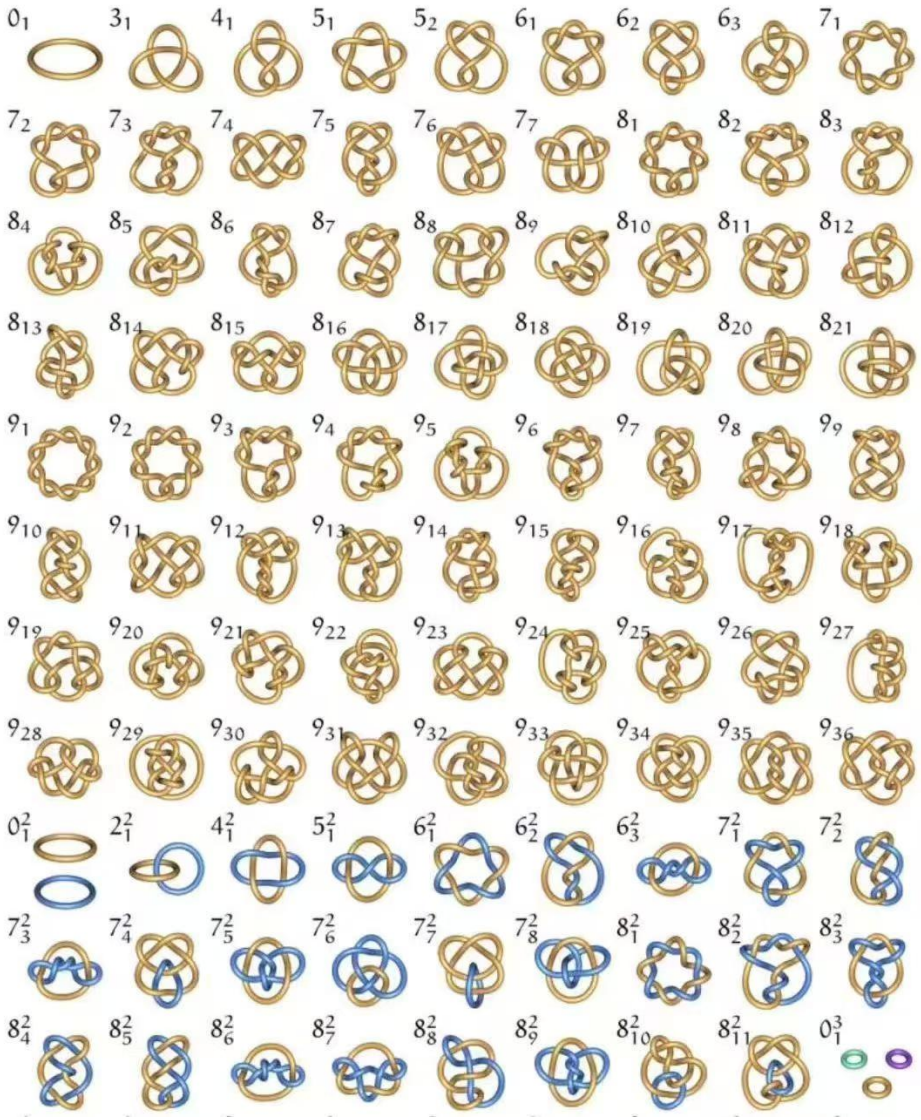


图 4: (Partial) Knot Classification Table

Knots And Links

Previous Invariant

The earliest invariants include **tricolorability** (which cannot distinguish knots with multiple crossings), **the crossing number of prime knots** (which cannot distinguish mirror-symmetric knots), **the unknotting number**, **linking number** and so on.However, these invariants have significant limitations.

Important method: The Skein Relation

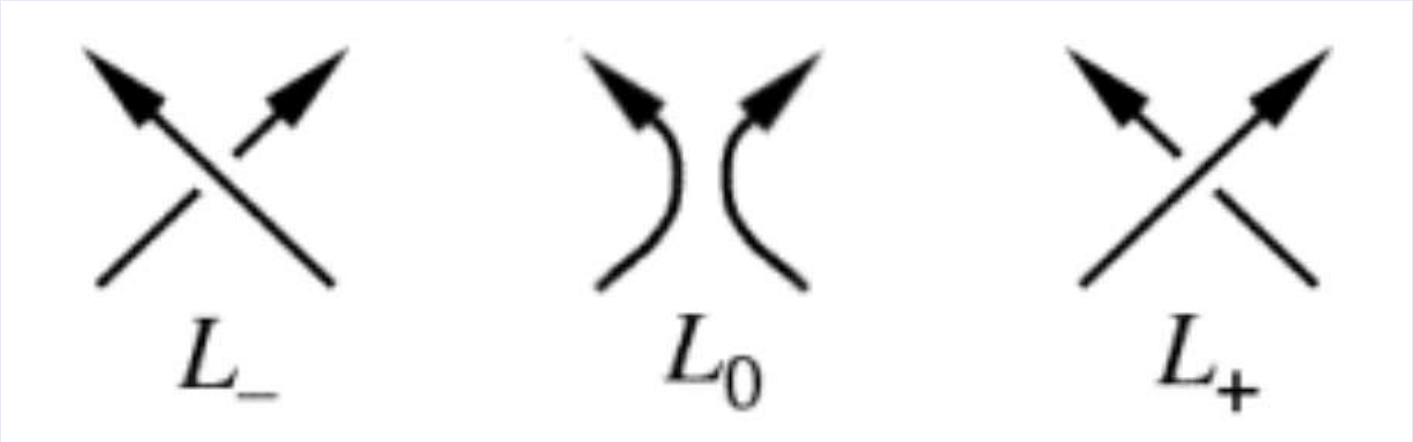


图 5: (Oriented Knot) Polynomial Invariant: $\mathbf{F(L_-, L_0, L_+) = 0}$

General Definition of the Alexander Polynomial

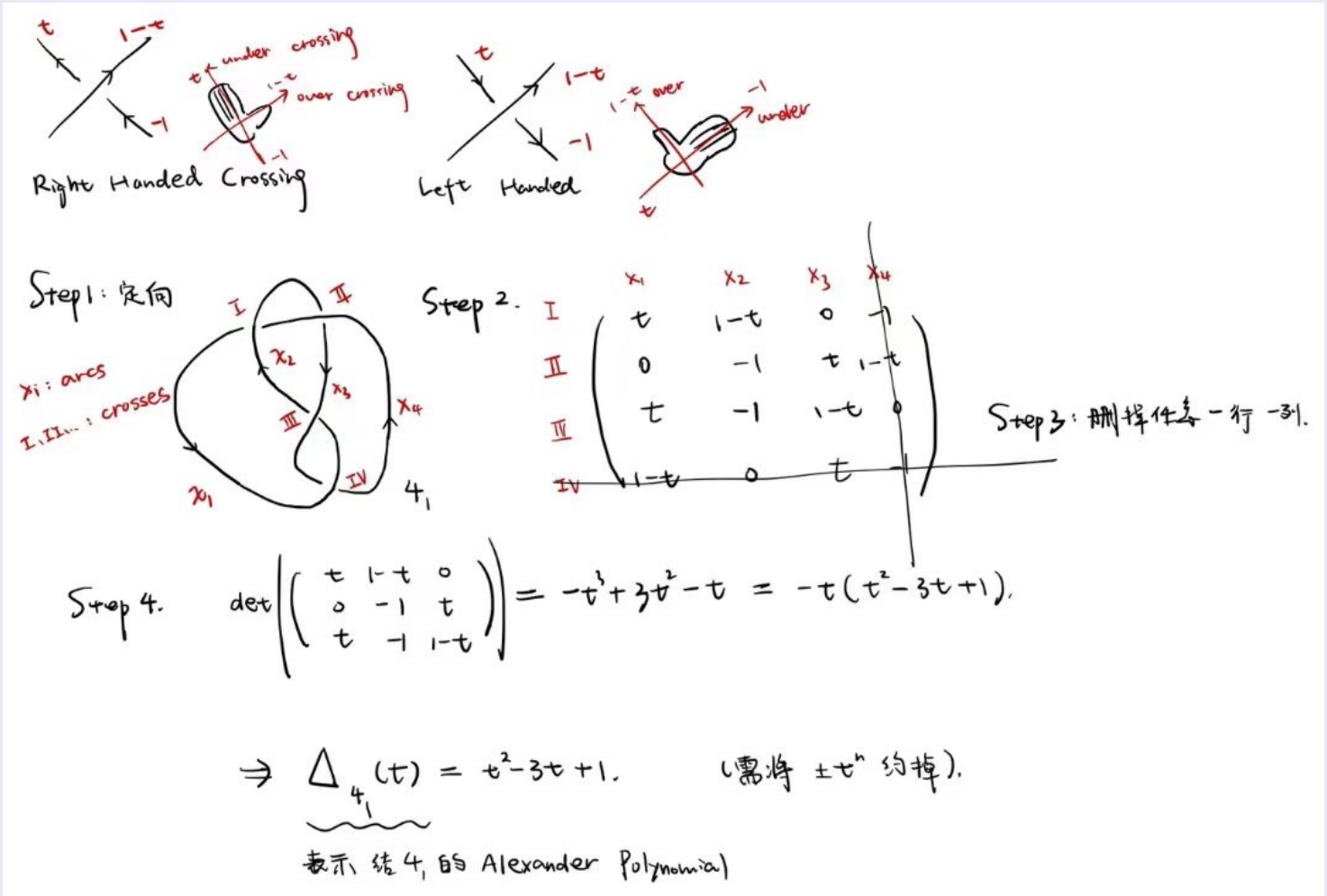


图 6: Picture from @7sato

Polynomial Invariant

Alexander Polynomial

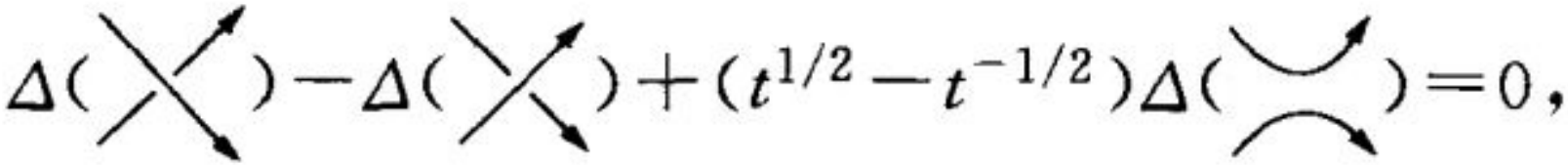


图 7: Relation Equation ($\triangle$(K) is Alexander Polynomial.)

Jones Polynomial

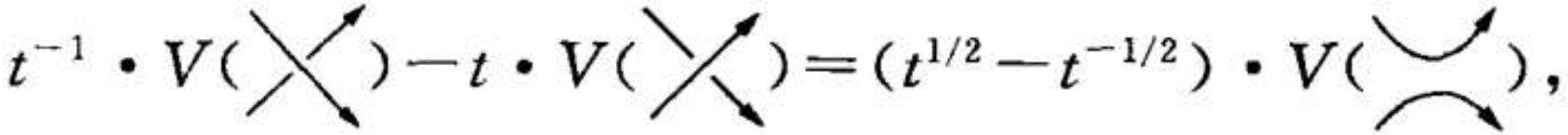


图 8: Relation Equation (V(K) is Jones Polynomial.)

Final Generalization: HOMFLY polynomial

$$P(\text{unknot}) = 1,$$

$$\ell P(L_+) + \ell^{-1} P(L_-) + mP(L_0) = 0,$$

$$P(L) = \frac{-(\ell + \ell^{-1})}{m} P(L_1) P(L_2).$$

图 9: The split union of two links L_1, L_2 of link L and the HOMFLY polynomial P(L).

Restriction

Relation

$$V(t) = P(\alpha = t^{-1}, z = t^{1/2} - t^{-1/2}),$$

$$\Delta(t) = P(\alpha = 1, z = t^{1/2} - t^{-1/2}),$$

图 10: The relationship among the three polynomial invariants.

Restriction of Alexander Polynomial

The Alexander polynomial can not detect chirality. For instan -ce, it can not distinguish between the left-handed trefoil knot and the right-handed trefoil knot.(Chirality, in simple terms, refers to mirror-image symmetry.)

Restriction of Jones Polynomial



Figure 11.2.3

图 11: The two knots have **the same Jones polynomial**:
$V(K_1) = V(K_2) = (t^{-2} - t^{-1} + 1 - t + t^2)^2$.

Restriction

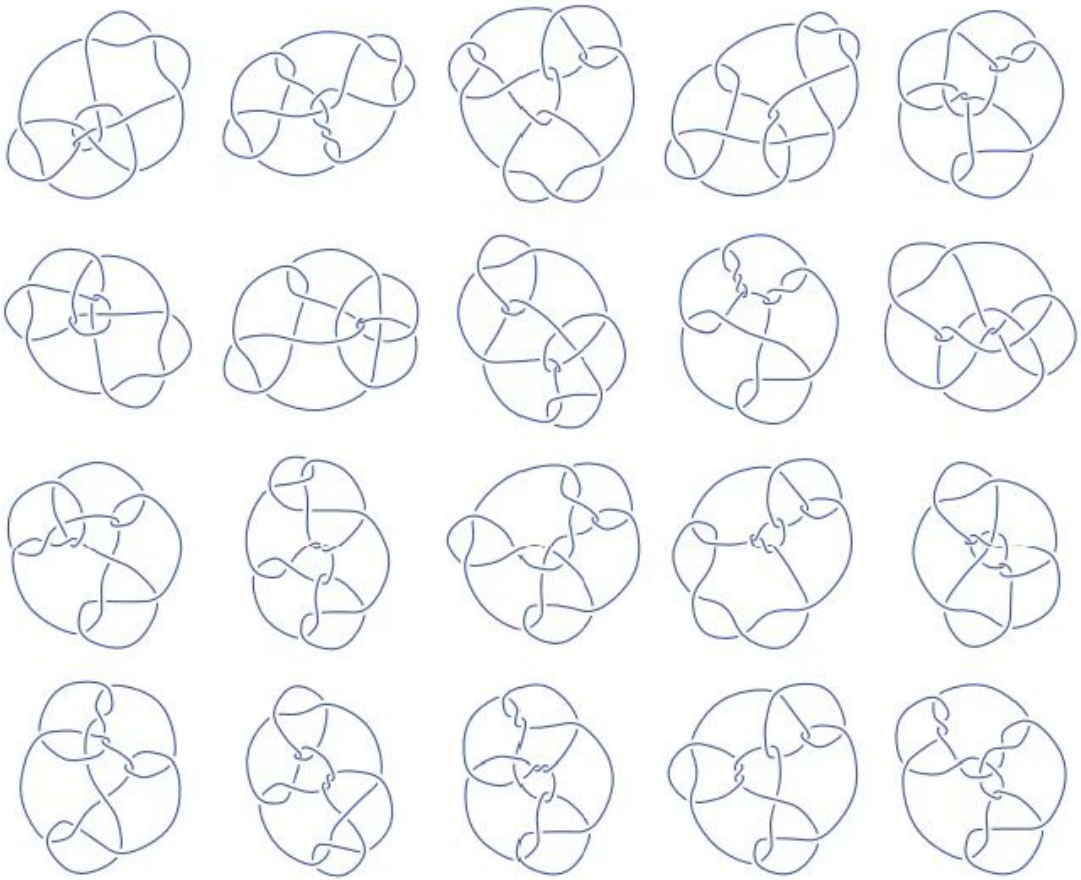


Figure 5: 20 distinct knots with the same Jones polynomial (from M.Thistlethwaite)

Categorification

Natural question

What is categorification, and why do we categorify invariants?

Ian Agol's comments on MathOverflow

Physicists view categorification as adding a time dimension in gauge theory.

The core of knot theory

The core of knot theory lies in classification, aiming to find complete knot invariants:
(1)It can distinguish more knots than existing invariants.
(2)It is easy to compute or represent.

Categorification

arXiv: 9802029

Categorification is the process of finding category-theoretic analogs of set-theoretic concepts by replacing sets with categories, functions with functors……

In brief

Categorification refines and improves the original incomplete invariants. It is not only a way to seek new and more complete invariants, but also a way to add/detect new layers of structure.

Categorification

(1) Euler Characteristic —— Singular homology
(2) Jones polynomial —— Khovanov homology
(3) Alexander Polynomial —— Knot Floer homology
(4) HOMFLY polynomial —— Khovanov-Rozansky homology.

Khovanov homology

Acknowledge

Assume here that you are familiar with chain complexes, boundary maps, and how to compute the homology group!

Smoothing of crossings

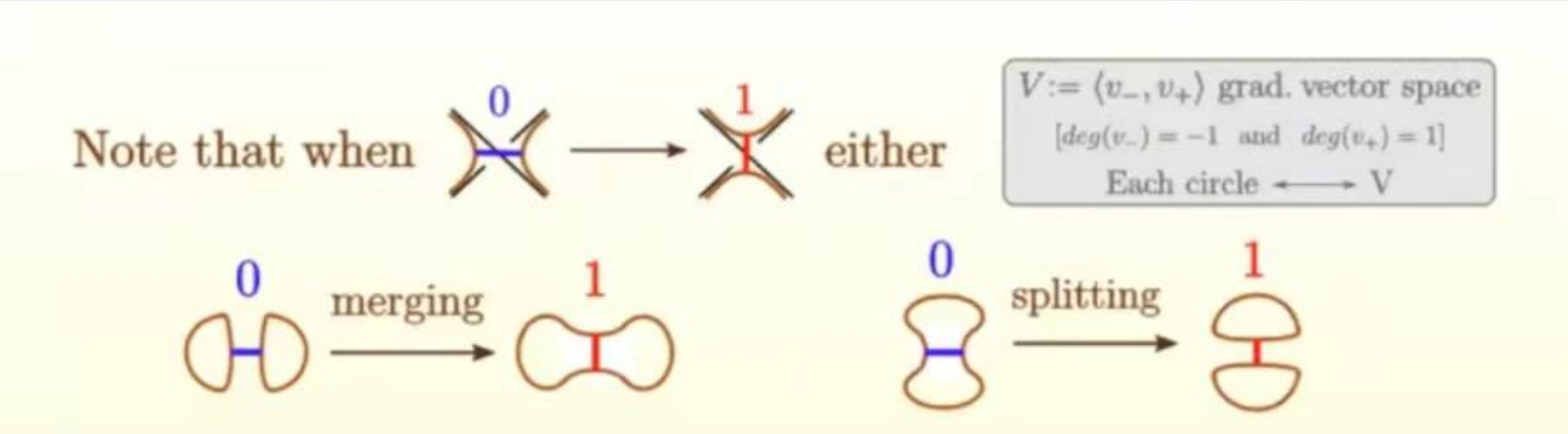


图 12: From the talk of Prof. Marithania Silvero

Method of processing

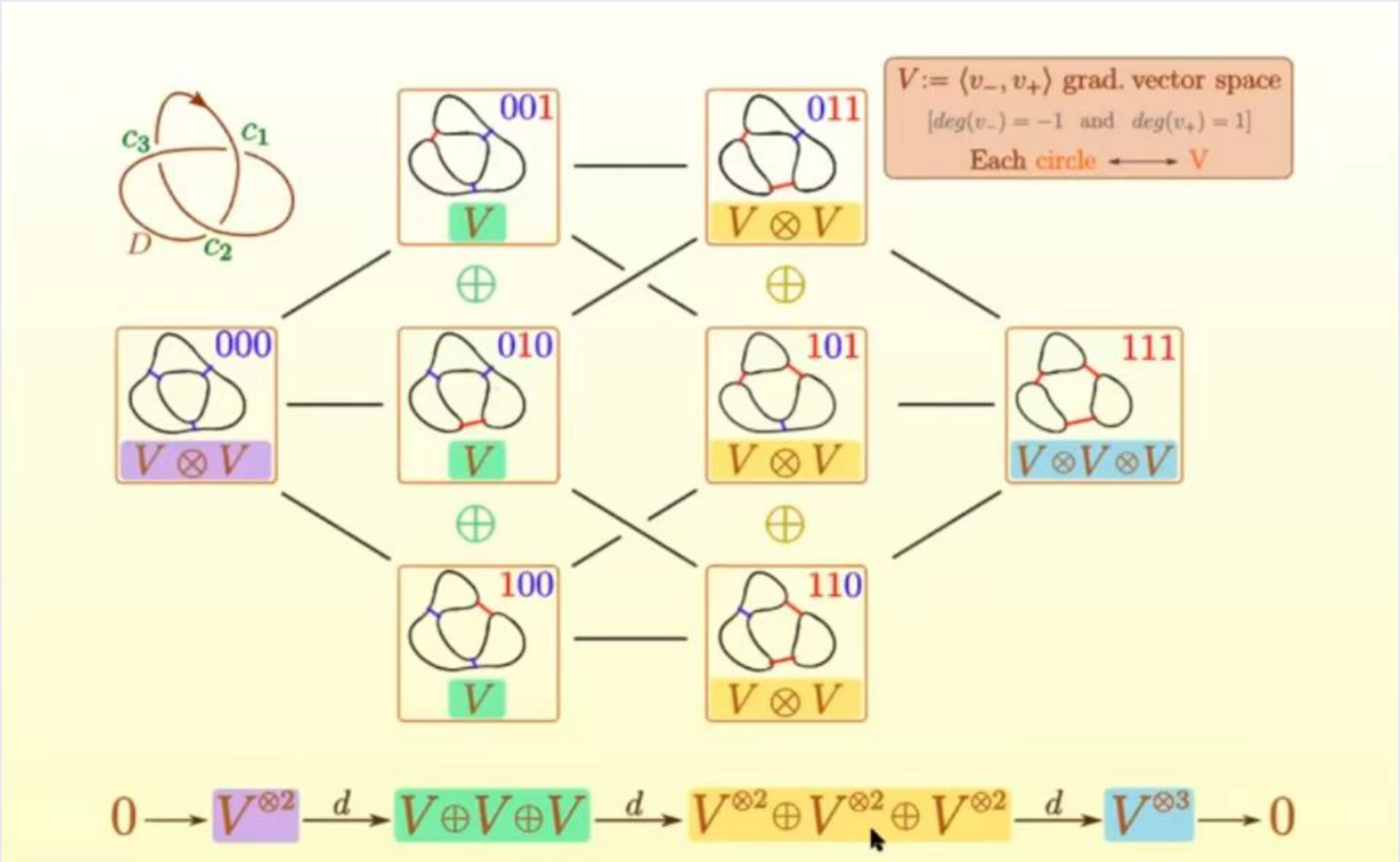


图 13: From the talk of Prof. Marithania Silvero.(Bracket States and Khovanov Complex)

Method of processing

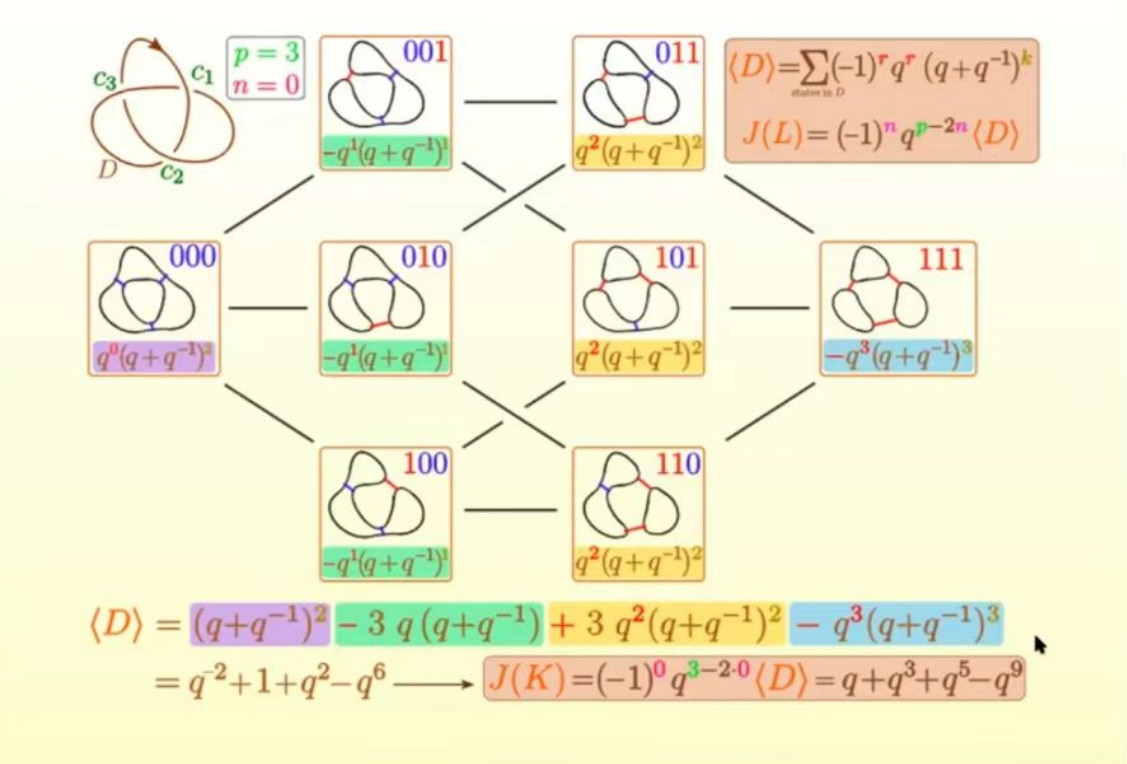


图 14: From the talk of Prof. Marithania Silvero. Each State is associated with Kauffman bracket and each crossing admits two state options.(r= number of 1-smoothing, k= number of circles)

Khovanov homology

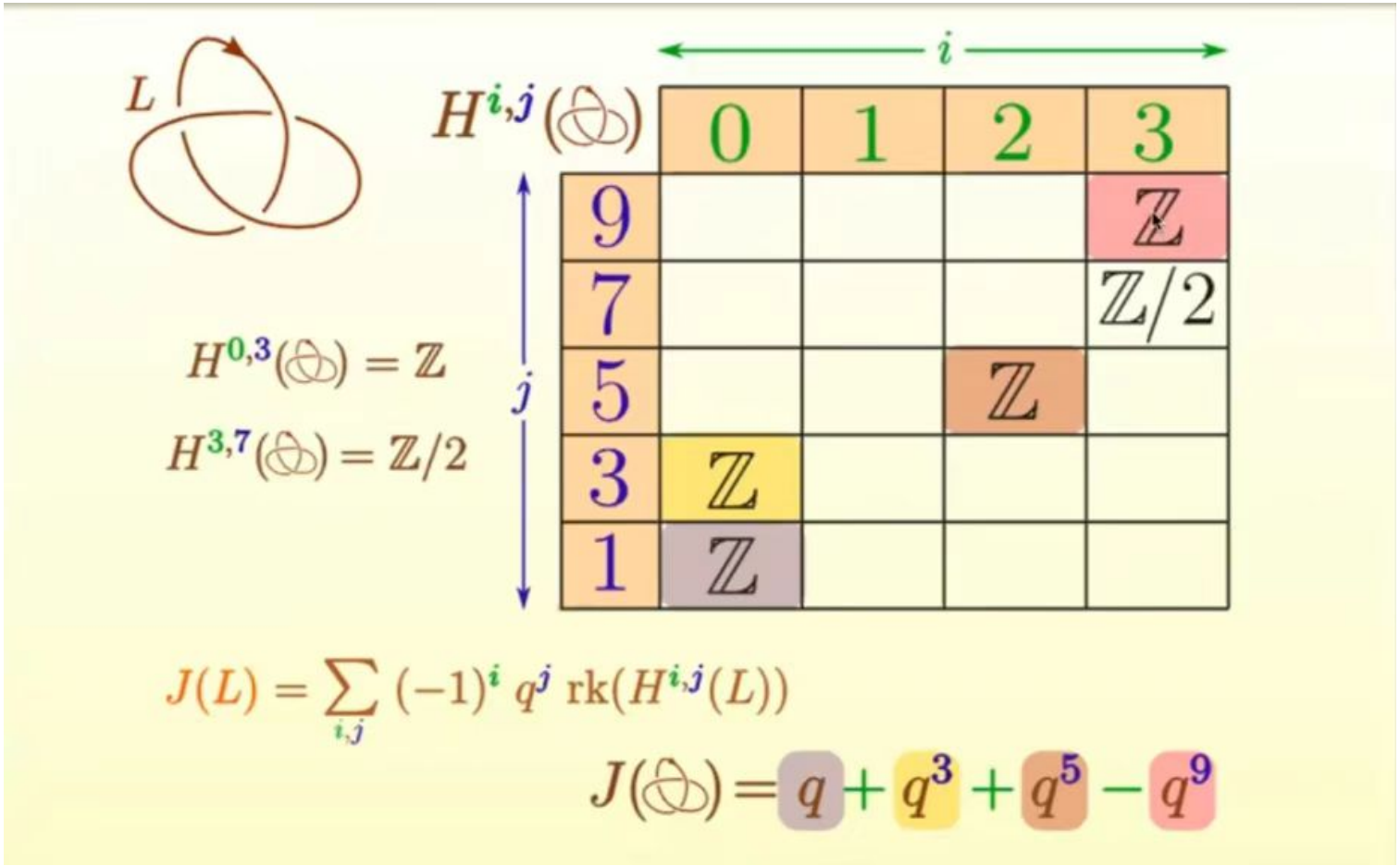


图 15: From the talk of Prof. Marithania Silvero. (i= number of 1-smoothing, j=(number of positive crossings)−(number of negative crossings)+(number of circles in S)−2×(number of 1-smoothings))

Khovanov homology

Remarks

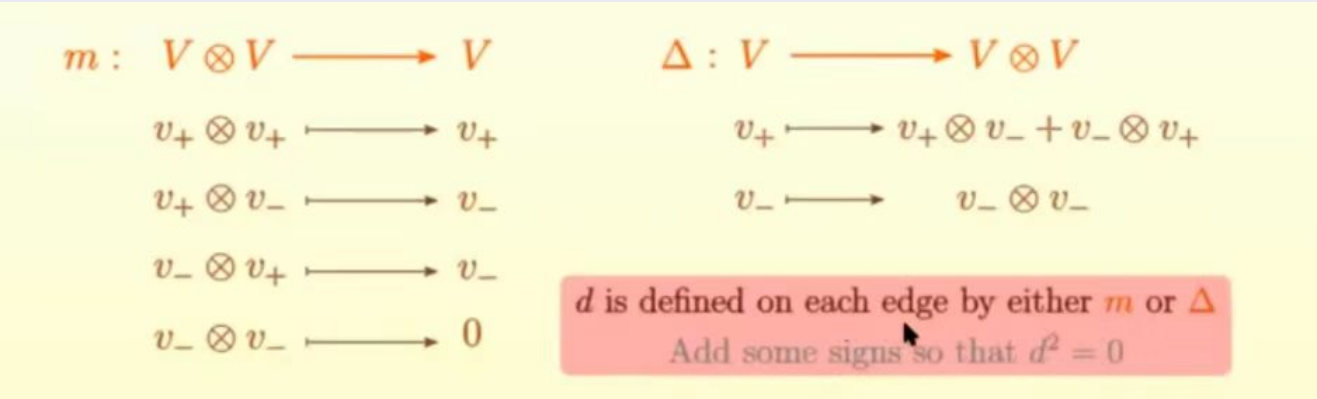


图 16: From the talk of Prof. Marithania Silvero.

Remarks

Among all products (bilinear maps), **the tensor product** preserves the full information of constituent components.

Khovanov homology

Supplement

(1)Khovanov homology is bi-graded, with each homology group specified by a pair of integer coordinates (i,j). Here i denotes the topological grading (Intuitively analogous to the dimension n in simplicial homology), and j stands for the quantum grading corresponding to the exponent of the polynomial.
(2)All chain spaces can be visualized as a two-dimensional grid. To compute the homology, one fixes j and computes separately with varying i. For a fixed j, a chain complex indexed by i is obtained, where each term $C^{i-1,j}$, $C^{i,j}$..... is a finite - dimensional vector space.

The most interesting construction

How is the boundary map ∂ in Khovanov homology constructed such that $\partial^2 = 0$?

Khovanov homology

Remark on Integral Differentials. Choose an ordering for the crossings in the link diagram K and denote them by $1, 2, \cdots n$. Let s be any enhanced state of K and let $\partial_i(s)$ denote the chain obtained from s by applying a partial boundary at the i-th site of s. If the i-th site is a smoothing of type A^{-1}, then $\partial_i(s) = 0$. If the i-th site is a smoothing of type A, then $\partial_i(s)$ is given by the rules discussed above (with the same signs). The compatibility conditions that we have discussed show that partials commute in the sense that $\partial_i(\partial_j(s)) = \partial_j(\partial_i(s))$ for all i and j. One then defines signed boundary formulas in the usual way of algebraic topology. One way to think of this regards the complex as the analogue of a complex in de Rham cohomology. Let $\{dx_1, dx_2, \cdots, dx_n\}$ be a formal basis for a Grassmann algebra so that $dx_i \wedge dx_j = -dx_j \wedge dx_i$ Starting with enhanced states s in $C^0(K)$ (that is, states with all A-type smoothings) define formally, $d_i(s) = \partial_i(s)dx_i$ and regard $d_i(s)$ as identical with $\partial_i(s)$ as we have previously regarded it in $C^1(K)$. In general, given an enhanced state s in $C^k(K)$ with B-smoothings at locations $i_1 < i_2 < \cdots < i_k$, we represent this chain as $s\, dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ and define

$$\partial(s\, dx_{i_1} \wedge \cdots \wedge dx_{i_k}) = \sum_{j=1}^{n} \partial_j(s)\, dx_j \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_k},$$

just as in a de Rham complex. The Grassmann algebra automatically computes the correct signs in the chain complex, and this boundary formula gives the original boundary formula when we take coefficients modulo two. Note, that in this formalism, partial differentials ∂_i of enhanced states with a B-smoothing at the site i are zero due to the fact that $dx_i \wedge dx_i = 0$ in the Grassmann algebra. There is more to discuss about the use of Grassmann algebra in this context. For example, this approach clarifies parts of the construction in [35].

图 17: From Louis H.Kauffman

Khovanov homology

It of interest to examine this analogy between the Khovanov (co)homology and de Rham cohomology. In that analogy the enhanced states correspond to the differentiable functions on a manifold. The Khovanov complex $C^k(K)$ is generated by elements of the form $s\, dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ where the enhanced state s has B-smoothings at exactly the sites $i_1, \cdots, i_k$. If we were to follow the analogy with de Rham cohomology literally, we would define a new complex $DR(K)$ where $DR^k(K)$ is generated by elements $s\, dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ where s is *any* enhanced state of the link K. The partial boundaries are defined in the same way as before and the global boundary formula is just as we have written it above. This gives a *new* chain complex associated with the link K. Whether its homology contains new topological information about the link K will be the subject of a subsequent paper.

In the case of de Rham cohomology, we can also look for compatible unitary transformations. Let M be a differentiable manifold and $\mathcal{C}(M)$ denote the DeRham complex of M over the complex numbers. Then for a differential form of the type $f(x)\omega$ in local coordinates $x_1, \cdots, x_n$ and ω a wedge product of a subset of $dx_1 \cdots dx_n$, we have

$$d(f\omega) = \sum_{i=1}^{n} (\partial f/\partial x_i) dx_i \wedge \omega.$$

Here d is the differential for the DeRham complex. Then $\mathcal{C}(M)$ has as basis the set of $|f(x)\omega\rangle$ where $\omega = dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ with $i_1 < \cdots < i_k$. We could achieve $Ud + dU = 0$ if U is a very simple unitary operator (e.g. multiplication by phases that do not depend on the coordinates x_i) but in general it will be an interesting problem to determine all unitary operators U with this property.

A further remark on de Rham cohomology. There is another relation with the de Rham complex: In [47] it was observed that Khovanov homology is related to Hochschild homology and Hochschild homology is thought to be an algebraic version of de Rham chain complex (cyclic cohomology corresponds to de Rham cohomology), compare [51].

图 18: From Louis H.Kauffman

Computation

Algorithm problem (算法问题)

参考：https://www.math.toronto.edu/ drorbn/papers/FastKh/
一般纽结的 Khovanov 同调群计算是相当复杂的，即便利用相关编程软件，计算速度很大程度上取决于 link diagram 的 girth (即一条水平线于 link diagram 相交的最大次数), 而每个固定 girth 的算法复杂度与交叉点的数量又呈低次多项式的关系。
“Girth 14 is critical and we can do up to about 80 crossings. Girth 12 and below is easy, meanwhile 100 crossings is plausible (合理的), but more likely in weeks. girth 16 and above seems to be out of range of recent computers. ”

Related Software and application

https://web.archive.org/web/20100830062946/http://www.math.uic.edu/ t3m/

Generalization

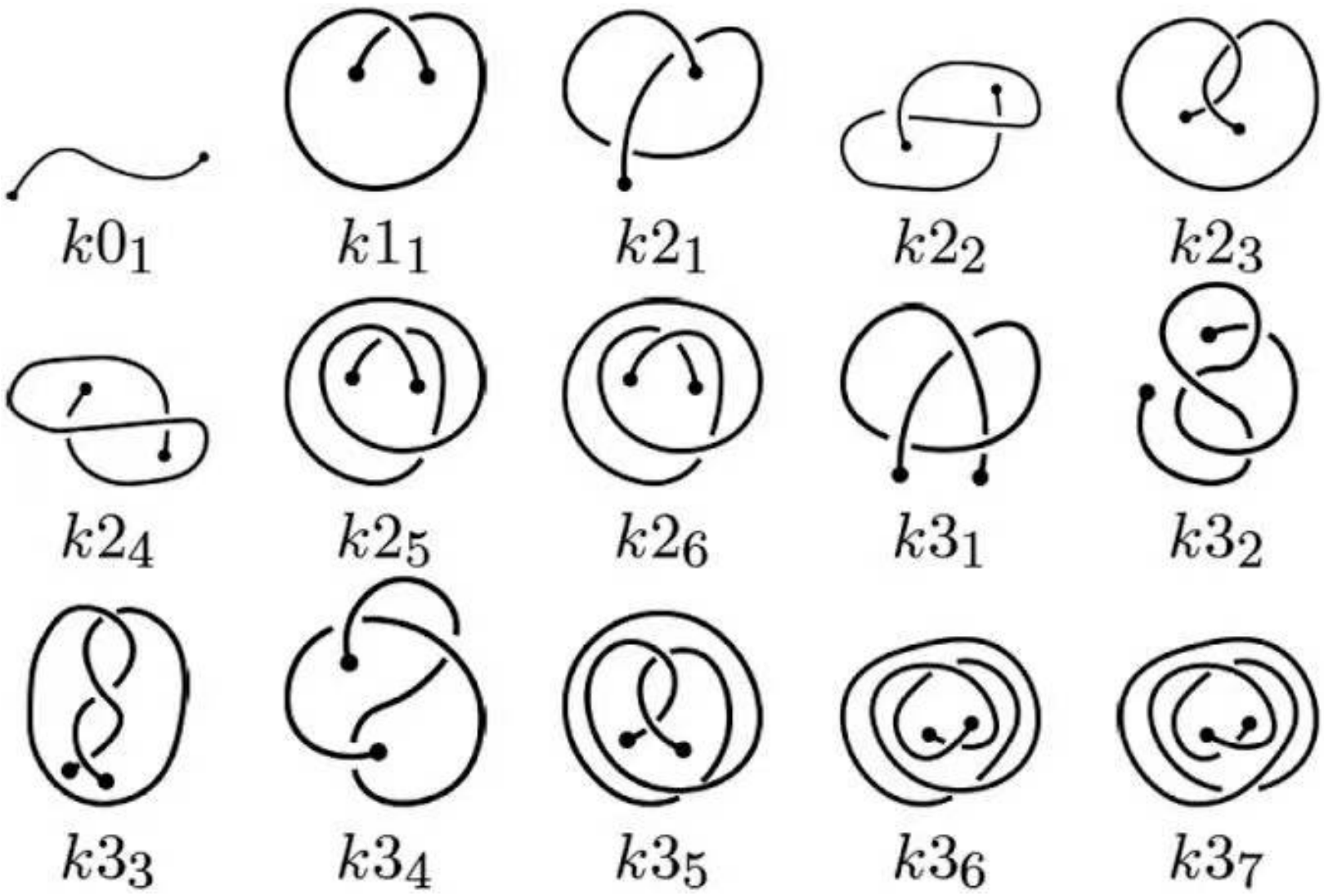


图 19: Knotoids

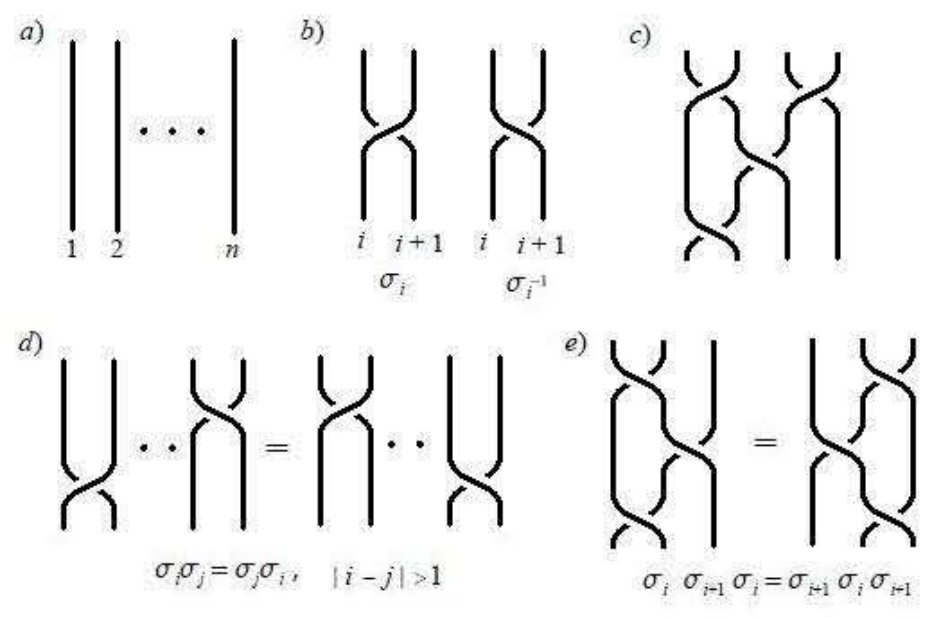
Knotoids

New Invariants

The odd writhe, the parity bracket polynomial,The affine index polynomial,The arrow polynomial.

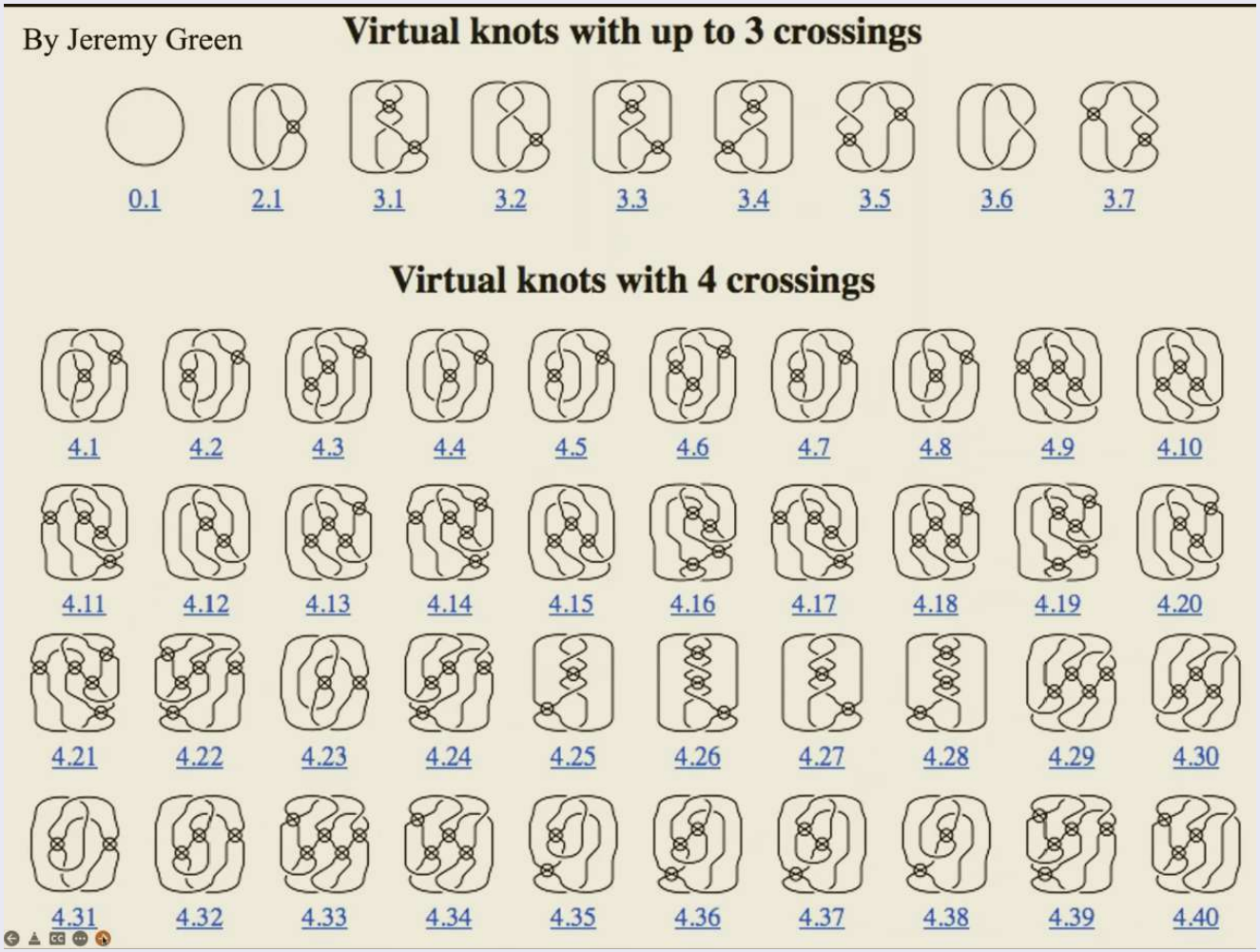
Reference： (1)arXiv:1002.4133 [Knotoids],V Turaev.
(2)arXiv:1602.03579 [New invariants of knotoids].

Comparison： Braids



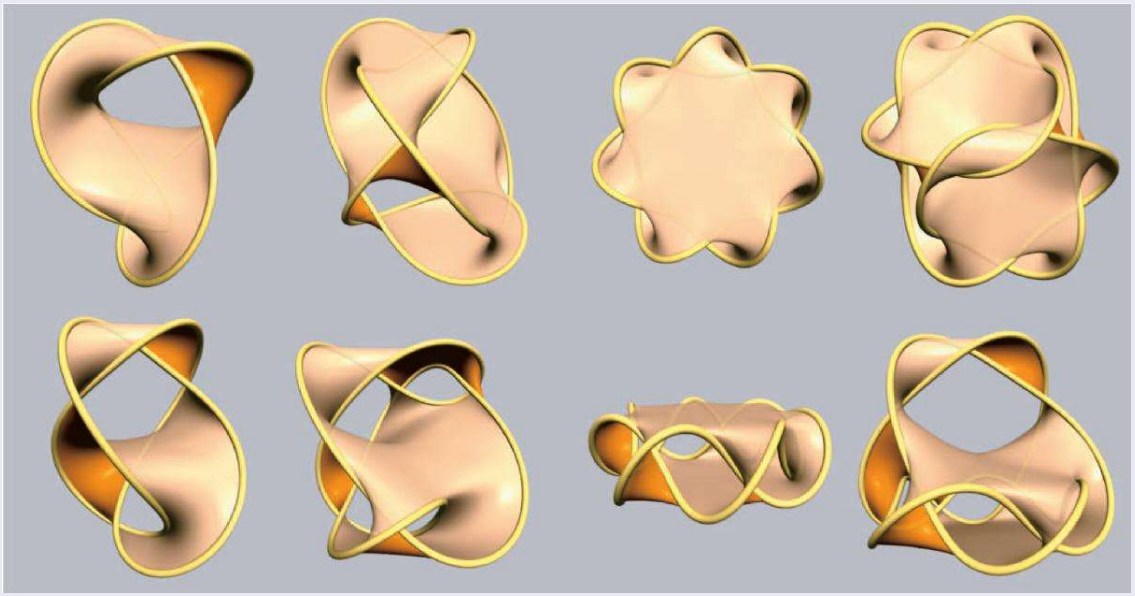
Virtual knot

(Kauffman,1999) Classical and Virtual crossings



Virtual knot

Seifert Genus (Important invariants)

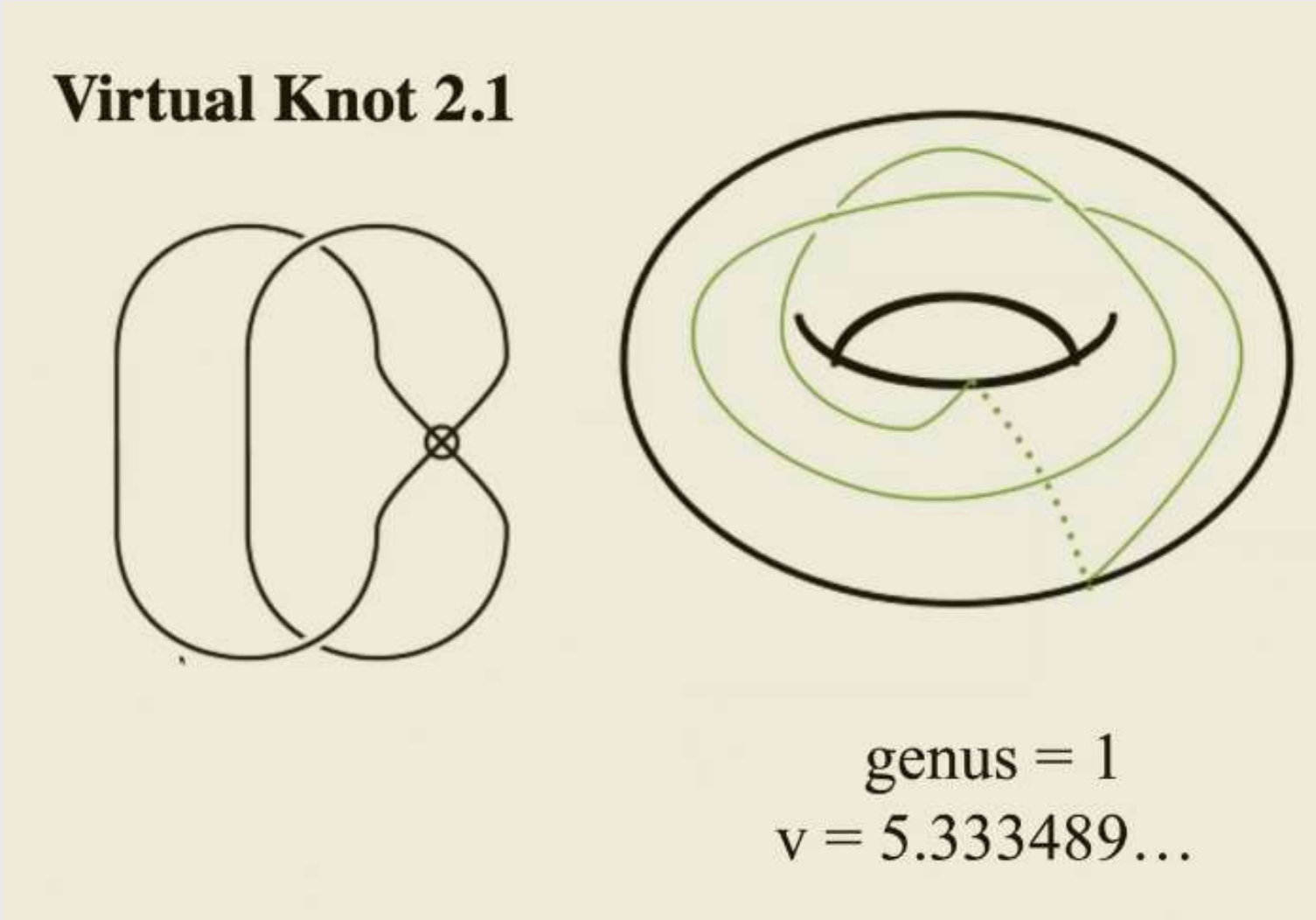


Construction

Every knot admits a compact oriented Seifert surface: Orient the knot, resolve every crossing to obtain a collection of simple closed curves, attach disks along these curves, and complete the construction by adding half-twisted band surfaces at each crossing.

Virtual knot

Genus and volume



Virtual knot

Virtual Knot	Minimal Genus	Hyperbolic Volume
2.1	1	5.33348956690
3.1	2	18.7531474071
3.2	1	7.70691180281
3.3	2	16.392265671 (same as 3.4)
3.4	2	16.392265671 (same as 3.3)
3.5	1	6.3545865572
3.6	0	Not hyperbolic
3.7	1	9.5034039310
4.1	2	22.493379859
4.2	2	22.611788156 (same as 4.69, 4.76, 4.98)
4.3	2	20.572042198 (same as 4.6)
4.4	2	21.200355468 (same as 4.30)
4.5	2	21.9154088043
4.6	2	20.572042198 (same as 4.3)
4.7	2	19.127715255
4.8	2	18.831683367 (same as 4.71, 4.77)
4.9	2	23.669963712
4.10	2	23.103877032
4.11	2	23.856837980
4.12	1	10.6669791338 (same as 4.53, 4.73, 4.75 and twice 2.1)
4.13	2	23.942021763
4.14	2	23.6657445099
4.15	2	19.472911306 (same as 4.35)
4.16	2	21.447588496
4.17	2	21.273940484
4.18	2	21.4281087698
4.19	2	21.384745444
4.20	2	17.9102390621 (same as 4.38, 4.50)
4.21	2	20.681258869
4.22	2	20.857960899
4.23	2	21.632850855
4.24	2	21.906855990
4.25	2	20.505494782 (same as 4.27)
4.26	2	21.301761314 (same as 4.28)
4.27	2	20.5054947824 (same as 4.25)
4.28	2	21.301761314 (same as 4.26)
4.29	2	21.961806381
4.30	2	21.200355468 (same as 4.4)
4.31	2	21.8298273713
4.32	2	21.551562929
4.33	2	21.710522971
4.34	2	22.071520116
4.35	2	19.472911306 (same as 4.15)
4.36	1	8.7933456039
4.37	1	6.5517432879
4.38	2	17.910239062 (same as 4.20, 4.50)
4.39	2	21.135859812
4.40	2	20.273838496 (same as 4.41)
4.41	2	20.2738384955 (same as 4.40)
4.42	2	21.328567359
4.43	1	8.9673608488
4.44	2	21.735522882 (same as 4.46)
4.45	2	23.626618267 (same as 4.47)
4.46	2	21.735522882 (same as 4.44)
4.47	2	23.6266182673 (same as 4.45)
4.48	2	20.618594305
4.49	2	21.2669639843
4.50	2	17.910239062 (same as 4.20, 4.38)
4.51	2	21.459015648 (same as 4.85)
4.52	2	21.456918384
4.53	1	10.6669791338 (same as 4.12, 4.73,4.75 and twice 2.1)
4.54	2	21.839438180
4.55	2	21.418632337 (**Kishino knot**)
4.56	2	23.773639799
4.57	2	18.9694297800
4.58	2	19.0560288254
4.59	2	18.831683367 (same as 4.8, 4.71, 4.77)
4.60	2	19.127715255 (same as 4.89)
4.61	2	21.8860118290 (same as 4.68)
4.62	2	23.5558331860
4.63	2	20.550520767
4.64	1	8.9293178231
4.65	1	10.5568662552
4.66	2	24.271905954
4.67	2	23.6009265895
4.68	2	21.886011829 (same as 4.61)
4.69	2	22.611788156 (same as 4.2,4.76, 4.98)
4.70	2	22.5937096032
4.71	2	18.831683367 (same as 4.8, 4.77)
4.72	2	22.161104637
4.73	1	10.6669791338 (same as 4.12, 4.53, 4.75 and twice 2.1)
4.74	2	21.1567287042
4.75	1	10.6669791338 (same as 4.12, 4.53, 4.73 and twice 2.1)
4.76	2	22.6117881556 (same as 4.2, 4.69, 4.98)
4.77	2	18.831833668 (same as 4.8, 4.71)
4.78	2	18.665545421 (same as 4.39, 4.79)
4.79	2	18.6655454521 (same as 4.39, 4.78)
4.80	2	17.593333761 (same as 4.81)
4.81	2	17.593333761 (same as 4.80)
4.82	2	19.498261755 (same as 4.84)
4.83	2	21.2165000389
4.84	2	19.498261755 (same as 4.82)
4.85	2	21.459015648 (same as 4.51)
4.86	1	8.7786588032
4.87	2	19.0663669549 (same as 4.88, 4.93)
4.88	2	19.0663669549 (same as 4.87, 4.93)
4.89	2	19.1277152546 (same as 4.60)
4.90	2	23.128377627
4.91	1	6.7551948165
4.92	1	7.51768989647
4.93	2	19.0663669549 (same as 4.87, 4.88)
4.94	1	9.9665118837
4.95	1	10.9616419532
4.96	2	22.698482328
4.97	2	24.053938791
4.98	2	22.611788156 (same as 4.2, 4.69, 4.76)
4.99	1	8.7385704088
4.100	1	8.3555021464
4.101	1	12.2113073079
4.102	1	10.714084294
4.103	2	24.604224640 (same as 4.88)
4.104	1	11.340719807
4.105	1	11.5188395851
4.106	1	12.0776477618
4.107	2	26.7015415469
4.108	0	2.02988321282

Staked Knots

(Definition) 补空间放置至少一个孤立点

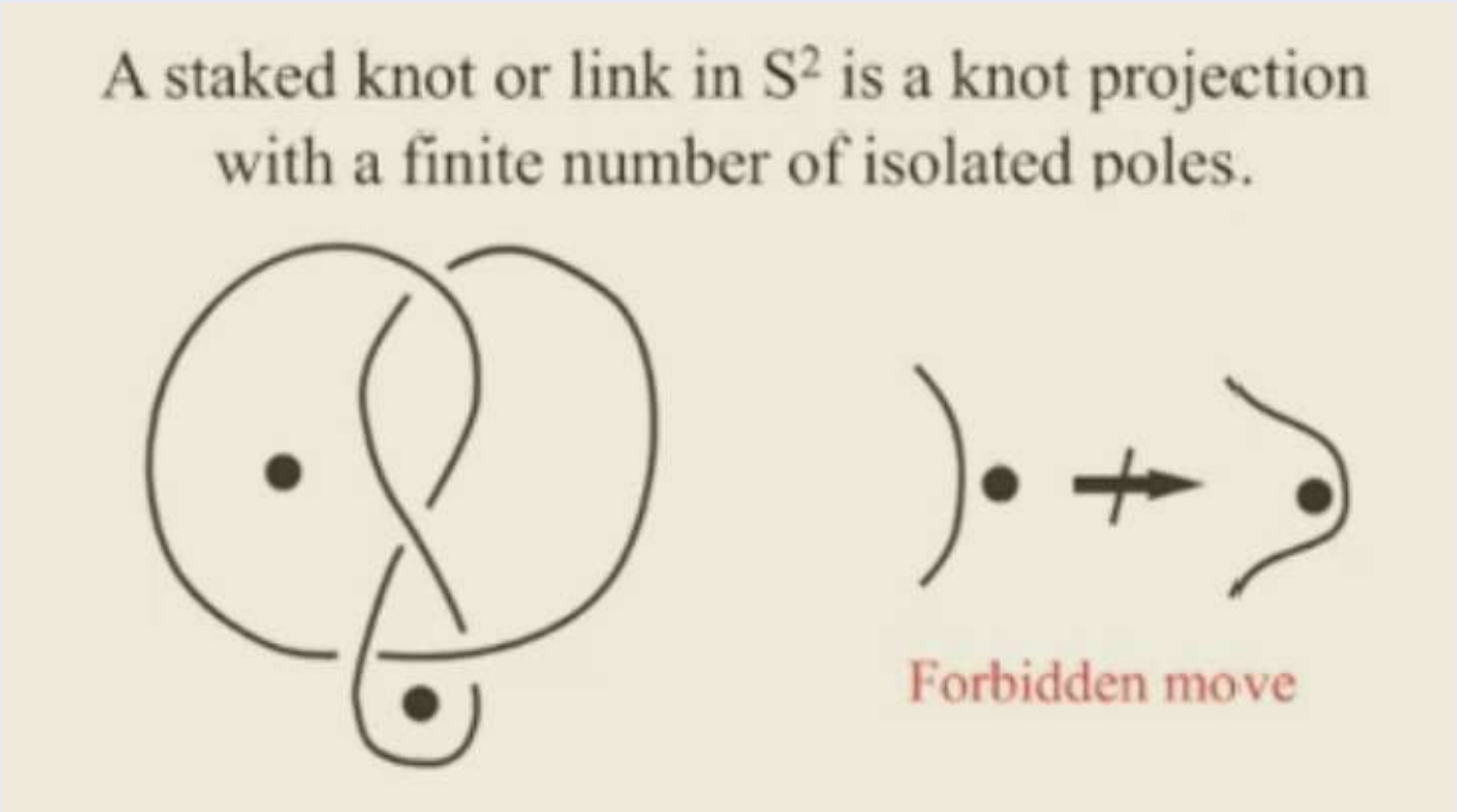
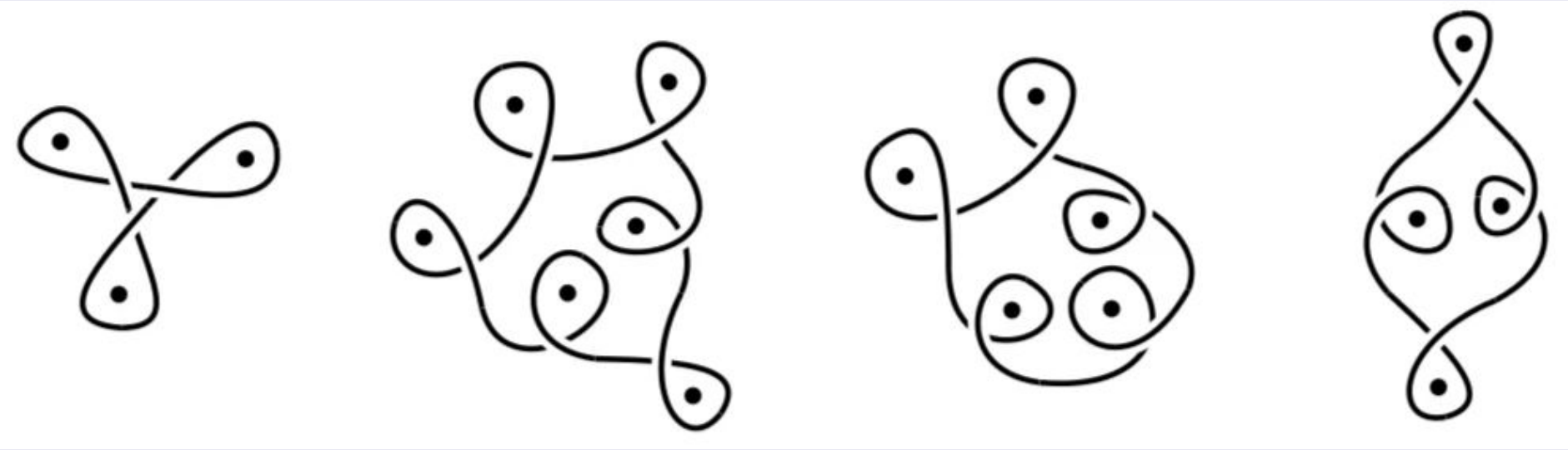


图 20: 研究这类特殊的纽结通常是看它的补空间的性质，比如论文 arXiv: 2603.00771, Hyperbolicity of staked links and lower bounds on their volumes.

Bongles

(Definition) 沿曲线遍历所有的极点



Property

如果曲线通过交叉点时呈现上下交替的规律，我们则称它为‘Alternating’，另外 **A bongle is hyperbolic if and only if it is alternating**. 参考论文 arXiv:2501.02360.

Bongles

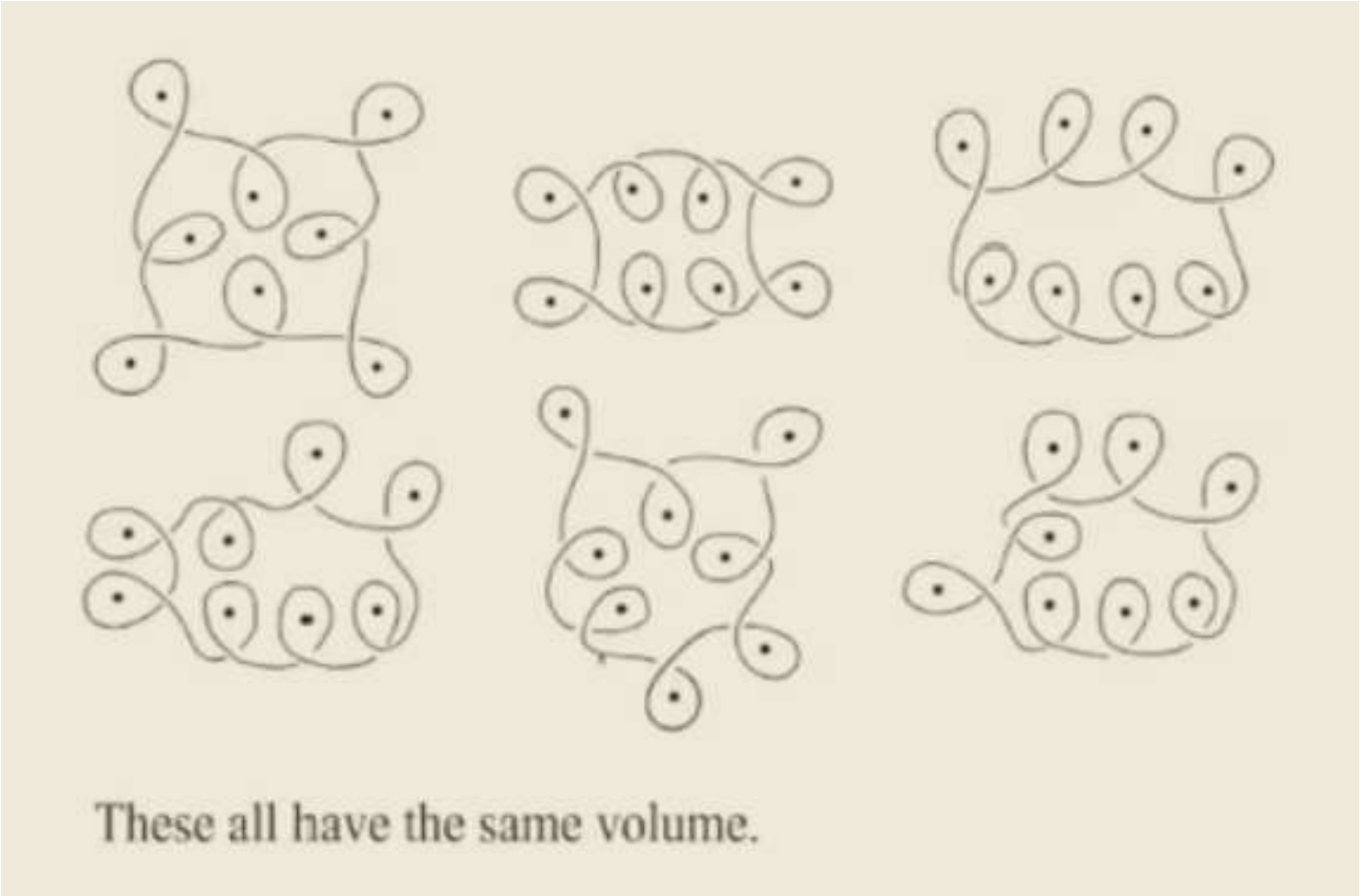


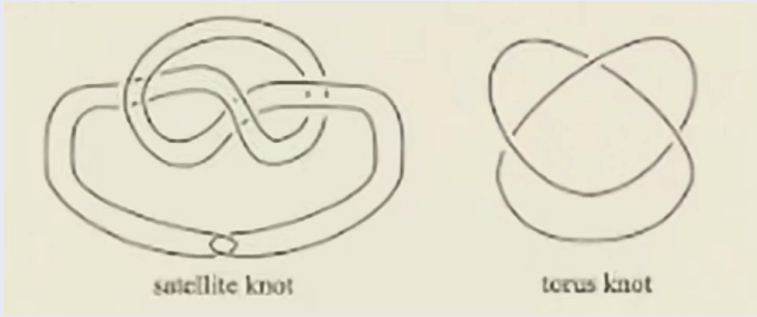
图 21: Switching innies to outies and vice versa (内旋结构与外旋结构) does change volume.

Connections with 3-/4-manifolds

Connections with 3-manifolds

The Fundamental Theorem of Dehn Surgery:
Every connected, orientable closed 3-manifold can be obtained by performing Dehn surgery on the 3-sphere S^3 along some link L. Therefore, the construction of common 3-manifolds is realized via the knot complement $S^3 \setminus K$ of a knot K in S^3. **Thurston discovered that the knot complement $S^3 \setminus K$ admits a hyperbolic structure if and only if K is neither a torus knot nor a satellite knot.**

Torus knot and satellite knot



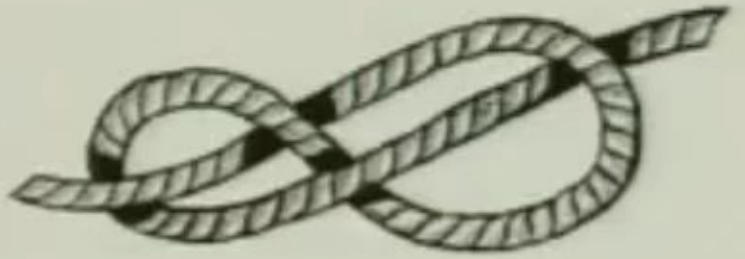
Important invariant of $S^3 \setminus K$

Hyperbolic volume

as a topological invariant

K =





$$\mathrm{vol}(S^3 - K) = 6\pi(\pi/3) = 6\int_0^{\pi/3} \log \frac{1}{2\sin\theta}\, d\theta$$

$$= 2.02988321281 93\ldots$$

Important invariant of $S^3 \setminus K$

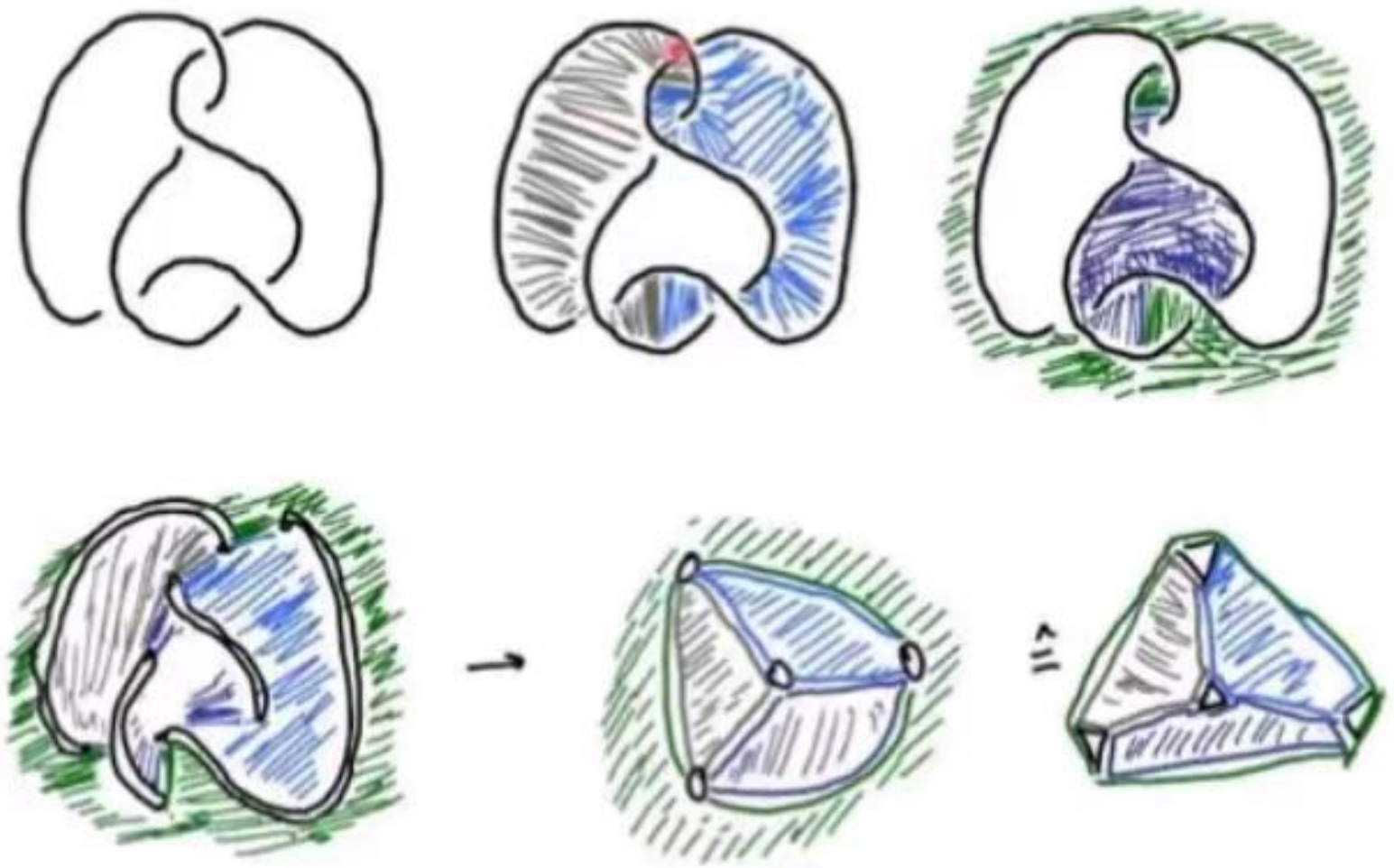


图 22: The knot complement $S^3 \setminus K$ is homeomorphic to two ideal hyperbolic tetrahedra glued together along their surfaces.

Important invariant of $S^3 \setminus K$



Volume Conjecture

Murakami-Murakami
Kashaev

$$V_{n+1}(K,t) = \sum_{j=0}^{n/2} (-1)^j \binom{n-j}{j} V(K^{n-2j}, t)$$

Cable K^2 for figure eight knot K

$$\frac{2\pi \ \log |V_n(K, e^{2\pi i/n})|}{n} \to \text{hyperbolic vol}(S^3\text{-}K)$$

quantum fields — general relativity

Connections with 4-manifolds

Topological slice/Smooth slice knot

A slice knot is a mathematical knot in 3-dimensional space that bounds an embedded disk in 4-dimensional space.A knot $K \subset S^3$ is said to be a topologically slice knot or a smoothly slice knot, **if it is the boundary of an embedded disk in the 4-ball B^4, which is locally flat or smooth, respectively.**

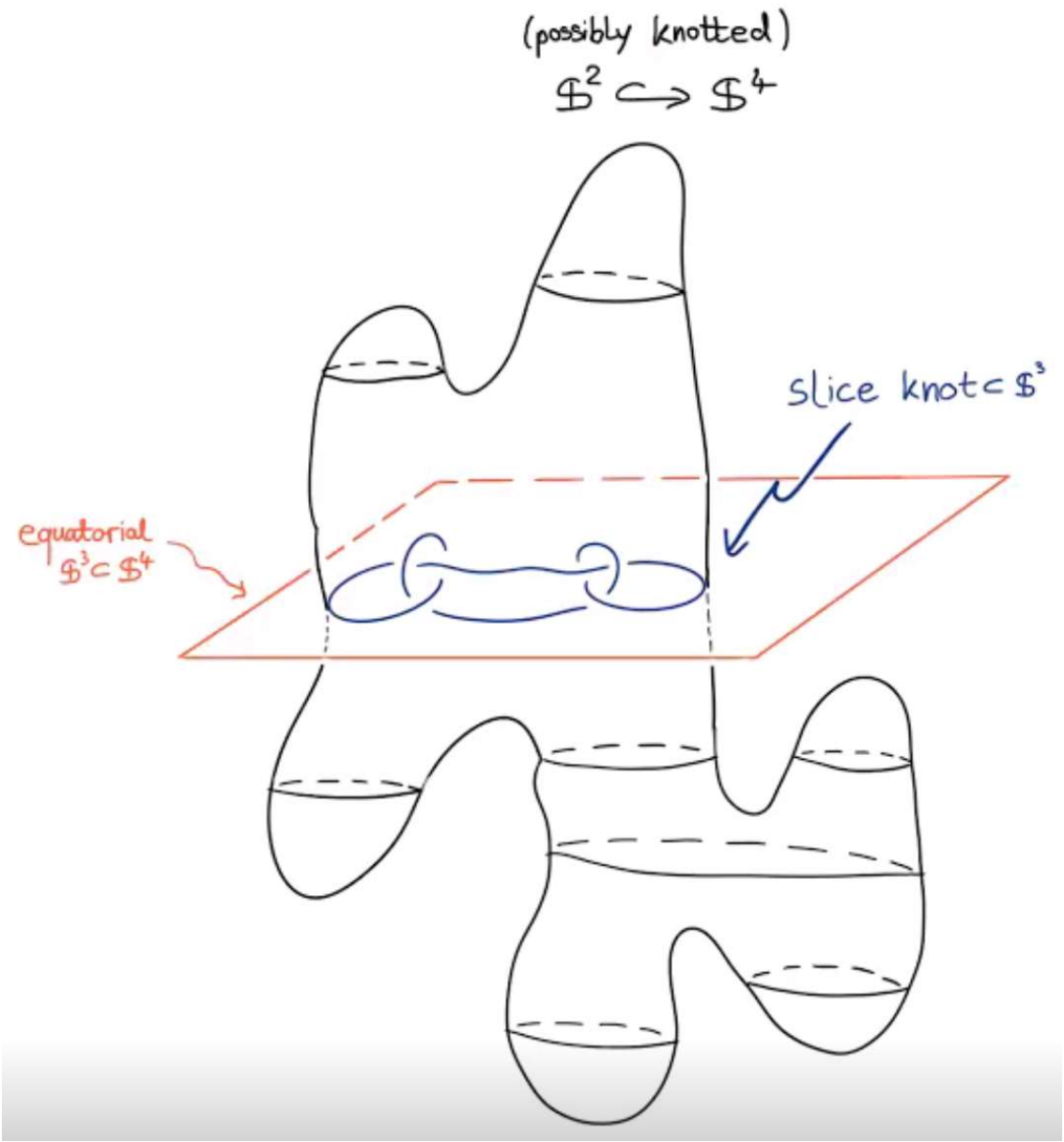
Freedman

If the Alexander polynomial of a knot is 1, then the knot is topologi -cally slice.

Construct exotic $\mathbb{R}^4$

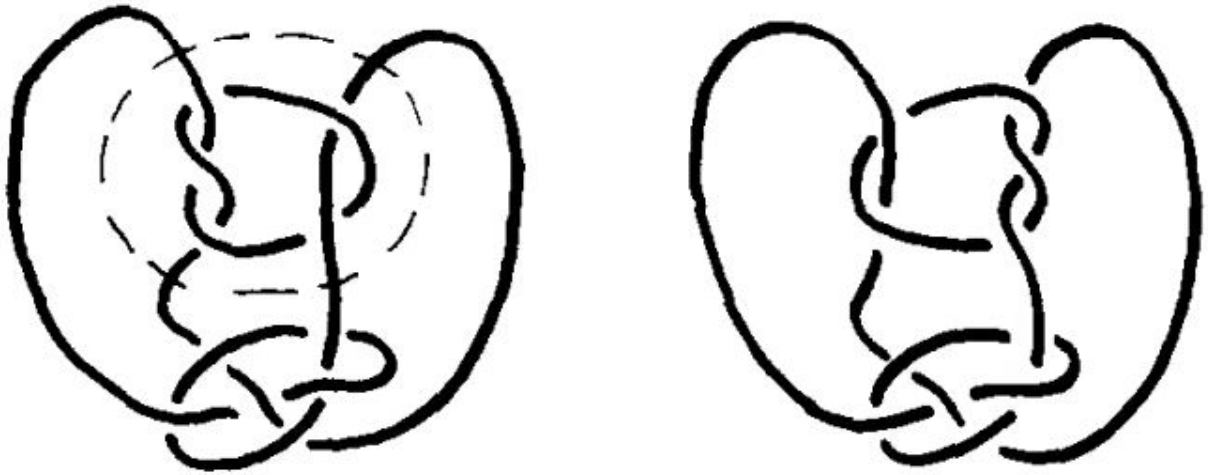
By the knot complement: if knot K is topologically slice not smooth slice, then the knot complement is topological embedding in R^4. **(How to Construct/find this knot?)**

Connections with 4-manifolds



Classic papers and recent progress

Lisa piccirillo: Conway knot is not slice. (arXiv:1808,02923, Annals.)



Famous mutants: the Conway and Kinoshita–Terasaka knots.

Proof sketch can be found: 【菲尔兹奖得主数学谈：拓扑学突破 | Lisa Piccirillo 如何证明康威纽结 (Conway Knot) 不是切片 (Slice) 纽结-哔哩哔哩】https://b23.tv/dWNQiIx

Unknotting number is not additive under connected sum. (arXiv:2506.24088)

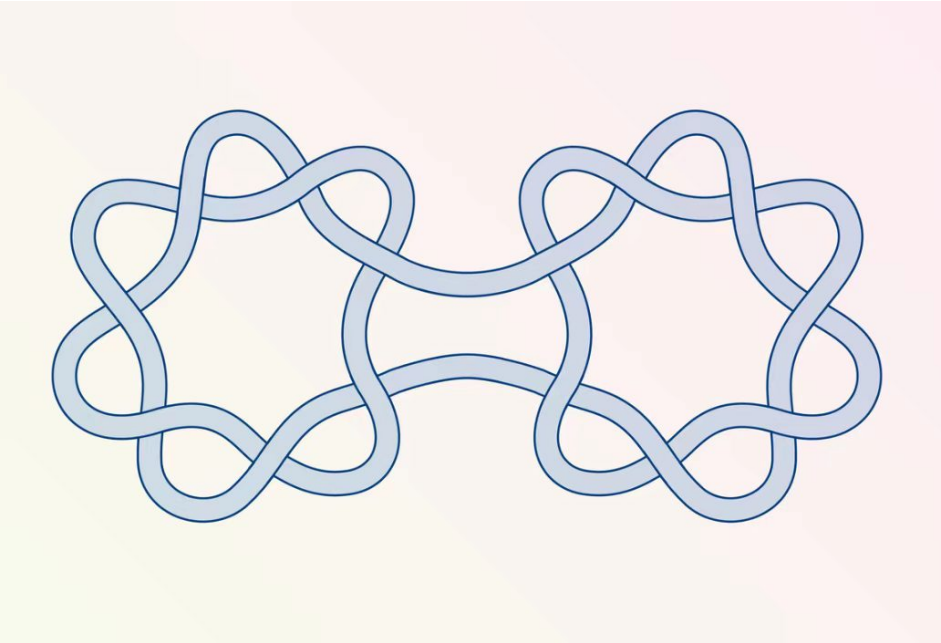
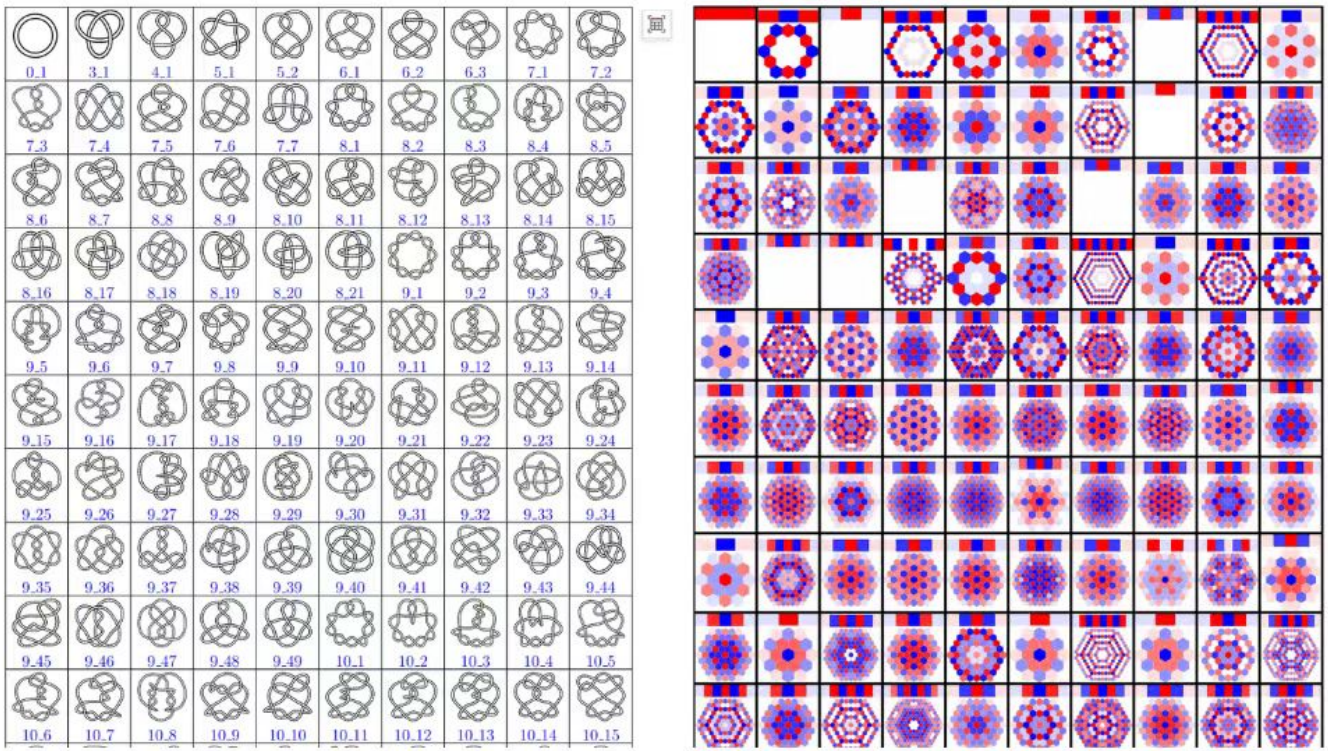


图 23: The connected sum of the (2,7)-torus knot and its mirror image.

Proof sketch can be found:【拓扑流形】几十年破解不了的猜想，答案竟在最简单的地方！谈谈前几天刚刚破解的扭结解结数猜想-哔哩哔哩】https://b23.tv/J8eCS5D

New knot invariants. (arXiv:2509.18456)



Proof sketch can be found: 【纽结也能扫码？-哔哩哔哩】
https://b23.tv/gAhMuGp (知乎 @ 三川啦啦啦)

Classic papers and recent progress

Several novel and significant papers on arXiv:

arXiv:2601.08767 如何用 Heegaard-floer 同调来区分 exotic $\mathbb{R}^4$.
arXiv:2507.15305 link homologies 与 knotted surface.
arXiv:2602.20138 Rasmussen s-invariants 与 exotic 4-manifold.
arXiv:1812.02448 The 4-dimensional Smale conjecture 被证伪
arXiv:2601.22152 4-流形中的 cobordism(配边-Manifold) 与 concordance(协边-Knot).
arXiv:math/0410606 如何通过 Spinning 方法来构造高维纽结.
……
(注：高维纽结的构造通常有两种方式：A.**Motives(可以理解为高维切片的方式)** B.**Spinning(纽结其中的一部分通过旋转)**).

Classic papers and recent progress

- c - the crossing number,
- c_3 - triple crossing index,
- c_Δ - the delta-crossing number,
- g_f - the free genus,
- g_c - the canonical genus,
- u - the unknotting number,
- u_b - the band-unknotting number,
- ul_b - the band-unlinking number,
- cl - the clasp number,
- td - the skein tree depth,
- tr - the trivializing number,
- br - the bridge number,
- b - the braid index,
- α - the arc index,
- m - the mosaic number,
- a - the ascending number.
- cl_4 - the 4D clasp number,
- g_4 - the slice genus,
- σ - the knot signature,
- g_{ds} - the doubly slice genus,
- g_r - the ribbon slice genus,
- u_r^* - the weak ribbon unknotting number,
- u_s - the slicing number,
- u_c - the concordance unknotting number.
- g - the genus,
- Ord_v - the torsion order,
- τ - the Ozsváth–Szabó's tau,
- s - the Rasmussen's s-invariant.
- $sp\Delta_t$ - the span of the Alexander polynomial $\Delta(t)$,
- spV_t - the span of the Jones polynomial $V(t)$,
- $degP_z$ - the z-degree of the HOMFLYPT polynomial $P_K(a,z)$,
- spP_a - the a-spread of the HOMFLYPT polynomial $P_K(a,z)$,
- spF_a - the a-spread of the Kauffman polynomial $F_K(a,z)$.

图 24: arXiv:2605.22652 ： some classic low-order invariants

Classic papers and recent progress

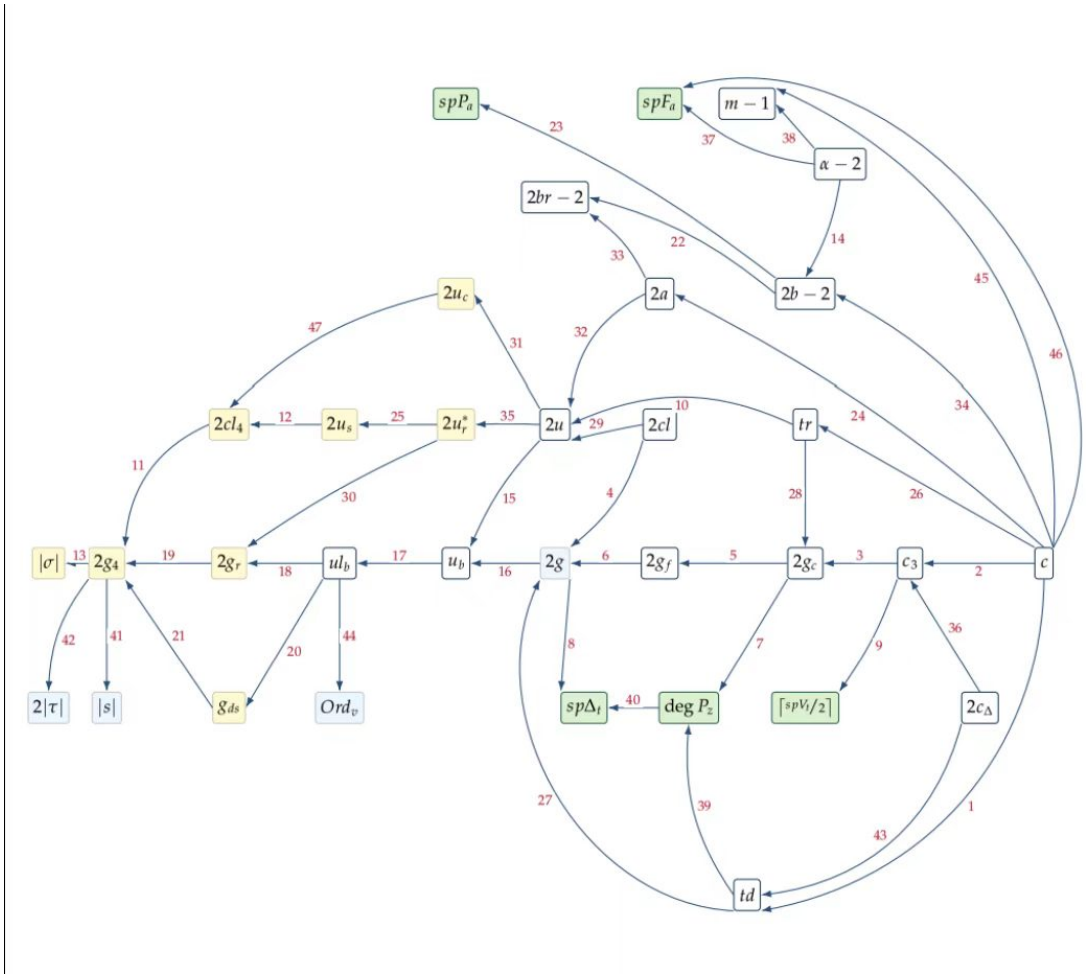


图 25: arXiv:2605.22652 ： some classic low-order invariants

(c1) $2br - 2 \leq 2cl$,
(c2) $2br - 2 \leq spF_a$,
(c3) $2br - 2 \leq tr$,
(c4) $2br - 2 \leq 2c_\Delta$,
(c5) $g_{ds} \leq 2u_r^*$,
(c6) $2g_f \leq 2cl$,
(c7) $2g_f \leq td$,
(c8) $2g_r \leq 2u_s$,
(c9) $2g_r \leq 2u_c$,
(c10) $2cl_4 \leq 2c_\Delta$.
(c11) $spP_a \leq spF_a$,
(c12) $\lceil spV_t/2 \rceil \leq td$,
(c13) $2|\tau| \leq degP_z$,
(c14) $|s| \leq degP_z$,
(c15) $|\sigma| \leq degP_z$,
(c16) $\alpha - 2 \leq c$,
(c17) $g_{ds} \leq 2u_c$,
(c18) $degP_z \leq 2cl$.

图 26: arXiv:2605.22652 ： some open problems

Classic papers and recent progress

The Maslov and Alexander functions induce a bigrading on $\widehat{\mathrm{CFK}}(\mathcal{H})$,

$$\widehat{\mathrm{CFK}}(\mathcal{H}) = \bigoplus_{d,s\in\mathbb{Z}} \widehat{\mathrm{CFK}}_d(\mathcal{H}, s),$$

where $\widehat{\mathrm{CFK}}_d(\mathcal{H}, s)$ is generated by those states $\mathbf{x}$ with $M(\mathbf{x}) = d$ and $A(\mathbf{x}) = s$. The differential satisfies

$$\widehat{\partial}_K : \widehat{\mathrm{CFK}}_d(\mathcal{H}, s) \to \widehat{\mathrm{CFK}}_{d-1}(\mathcal{H}, s),$$

and therefore the bigrading descends to homology

$$\widehat{\mathrm{HFK}}(\mathcal{H}) = \bigoplus_{d,s\in\mathbb{Z}} \widehat{\mathrm{HFK}}_d(\mathcal{H}, s).$$

The bigraded chain complex $\widehat{\mathrm{CFK}}(\mathcal{H})$ has a *graded Euler characteristic*, which is a Laurent polynomial with integral coefficients, in a formal variable t, defined by

$$\chi(\widehat{\mathrm{CFK}}(K)) = \sum_{d,s} (-1)^d \dim \widehat{\mathrm{CFK}}_d(K, s) t^s = \sum_{d,s} (-1)^d \dim \widehat{\mathrm{HFK}}_d(K, s) t^s.$$

This graded Euler characteristic coincides with the Alexander polynomial $\Delta_K(t)$:

$$\chi(\widehat{\mathrm{CFK}}(K)) \doteq \Delta_K(t), \tag{2.1}$$

where here $\doteq$ means that the two polynomials agree up to overall factors of t [91].

图 27: Knot floer homology

Classic papers and recent progress

Theorem 4.2. [90] *Knot Floer homology detects the Seifert genus $g(K)$ of a knot K, in the sense that*

$$g(K) = \min\{s | \widehat{\mathrm{HFK}}_*(K, s) \neq 0\}.$$

Our original proof of the above theorem relied on many results in low-dimensional topology. To start with, Gabai's theory of sutured manifolds equips the zero-surgery with a taut foliation [37]. A theorem of Eliashberg and Thurston [24] provides the product $[-1, 1] \times S^3_0(K)$ with a symplectic structure, which is suitably convex at the boundary. A theorem of Eliashberg [23] and Etnyre [25] embeds this symplectic cylinder in a closed symplectic manifold X. Donaldson's Lefschetz pencils [18] on symplectic manifolds then provides a suitable two-handle decomposition on X for which we can prove that the Heegaard Floer four-manifold invariant is non-zero [93]. A surgery formula relating knot Floer homology with the Heegaard Floer homology of the 0-surgery then gives the required non-vanishing theorem for knot Floer homology.

Juhász has an elegant proof of the above result [45] that bypasses most of the above machinery (still building on Gabai's sutured hierarchy), using his *sutured Floer homology* [44].

Theorem 4.2 has the following corollary:

Corollary 4.3. [90] *Knot Floer homology detects the unknot, in the sense that $\widehat{\mathrm{HFK}}(K)$ has dimension one if and only if K is the unknot.*

The above corollary underscores how far knot Floer homology goes beyond the Alexander polynomial: there are infinitely many knots with trivial Alexander polynomial.

Theorem 4.1 has a more precise statement, which expresses the sequence $\{\alpha_k\}$ concretely in terms of the surgery coefficient r and the resulting lens space $L(p, q)$ [94]. In [90], we combine this result with Corollary 4.3, to obtain the following result, first proved using Seiberg-Witten theory in our joint work with Kronheimer and Mrowka:

Corollary 4.4. [59] *If $K \subset S^3$ is a knot with the property that $S^3_r(K) \cong \mathbb{RP}^3$, then K is the unknot.*

See [41] for a vast generalization.

A final property of knot Floer homology motivated by the Alexander polynomial is based on the classical result that the Alexander polynomial of a fibered knot is monic. This has an analogue for knot Floer homology: if K is a fibered knot with Seifert genus $g = g(K)$, then

$$\widehat{\mathrm{HFK}}_*(K, g) = \bigoplus_{d \in \mathbb{Z}} \widehat{\mathrm{HFK}}_d(K, g)$$

is one-dimensional [93]. This fact has the following remarkable converse, due to Paolo Ghiggini when $g = 1$ and Yi Ni when $g > 1$:

Theorem 4.5. [38, 82] *If $\widehat{\mathrm{HFK}}_*(K, g)$ is one-dimensional, then K is fibered.*

See also [45].

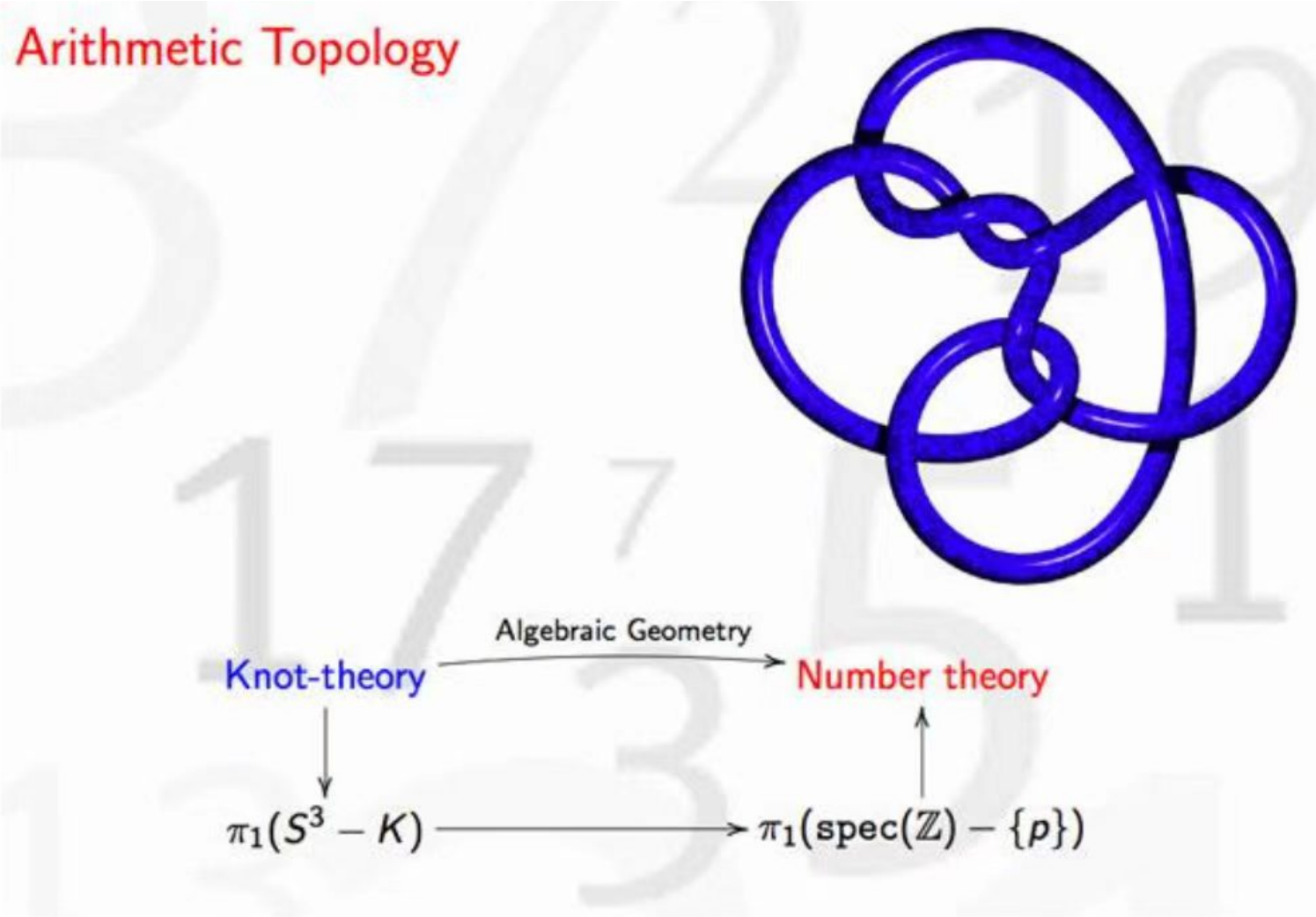
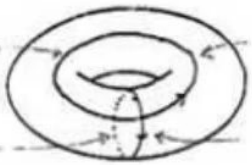

Arithmetic Topology
Algebraic Geometry
Knot-theory
Number theory
$\pi_1(S^3 - K)$
$\pi_1(\mathrm{spec}(\mathbb{Z}) - \{p\})$

图 28: 算术拓扑

Classic papers and recent progress

number ring $\mathrm{Spec}(\mathcal{O}_k) \cup \{\infty\}$ $\mathrm{Spec}(\mathbf{Z}) \cup \{\infty\}$	$\longleftrightarrow$	3-manifold M S^3
prime : $\mathrm{Spec}(\mathbf{F}_\mathfrak{p}) \subset \mathrm{Spec}(\mathcal{O}_k)$ primes $\mathfrak{p}_1, \cdots, \mathfrak{p}_n$ infinite prime	$\longleftrightarrow$	knot $K : S^1 \subset M$ link $K_1 \cup \cdots \cup K_n$ end
$\mathfrak{p}$-adic integers $\mathrm{Spec}(\mathcal{O}_\mathfrak{p})$	$\longleftrightarrow$	tube n.b.d $V(K)$
$\mathfrak{p}$-adic field $\mathrm{Spec}(k_\mathfrak{p})$	$\longleftrightarrow$	torus $\partial V(K)$
$\pi_1(\mathrm{Spec}(\mathcal{O}_\mathfrak{p})) = \langle \sigma \rangle$	$\longleftrightarrow$	$\pi_1(V(K)) = \langle \beta \rangle$
$\pi_1^{tame}(\mathrm{Spec}(k_\mathfrak{p})) = \langle \tau, \sigma \mid \tau^{p-1}[\tau, \sigma] = 1 \rangle$	$\longleftrightarrow$	$\pi_1(\partial V(K)) = \langle \alpha, \beta \mid [\alpha, \beta] = 1 \rangle$
σ : Frobenius auto. τ : monodromy		 β : longitude α : meridian
$k^\times \to \bigoplus_{\mathfrak{p}:\text{primes}} \mathbf{Z}$ $a \mapsto a\mathcal{O}_k$	$\longleftrightarrow$	$C_2(M, \mathbf{Z}) \xrightarrow{\partial} C_1(M, \mathbf{Z})$ $\Sigma \mapsto \partial\Sigma$
class group H_k	$\longleftrightarrow$	$H_1(M, \mathbf{Z})$
$\mathcal{O}_k^\times$	$\longleftrightarrow$	$H_2(M, \mathbf{Z})$
$\pi_1(\mathrm{Spec}(\mathcal{O}_k) \setminus \{\mathfrak{p}_1, \cdots, \mathfrak{p}_n\})$ max. Galois group with given ramification	$\longleftrightarrow$	$\pi_1(M \setminus K_1 \cup \cdots \cup K_n)$ link group
power residue symbol	$\longleftrightarrow$	linking number

Classic papers and recent progress

oriented, connected, closed 3-manifold $\overline{M}$	compactified spectrum of number ring $\overline{X}_k = \overline{\mathrm{Spec}(\mathcal{O}_k)}$
knot $\mathcal{K} : S^1 \hookrightarrow M$	prime $\{\mathfrak{p}\} = \mathrm{Spec}(\mathcal{O}_k/\mathfrak{p}) \hookrightarrow \overline{X}_k$
link $\mathcal{L} = \mathcal{K}_1 \sqcup \cdots \sqcup \mathcal{K}_r$	finite set of maximal ideals $S = \{\mathfrak{p}_1, \ldots, \mathfrak{p}_r\}$
tubular n.b.d of a knot $V_{\mathcal{K}}$ boundary torus $\partial V_{\mathcal{K}}$ peripheral group $\pi_1(\partial V_{\mathcal{K}})$	$\mathfrak{p}$-adic integer ring $V_{\mathfrak{p}} = \mathrm{Spec}(\mathcal{O}_{\mathfrak{p}})$ $\mathfrak{p}$-adic field $\partial V_{\mathfrak{p}} = \mathrm{Spec}(k_{\mathfrak{p}})$ local absolute Galois group $\Pi_{\mathfrak{p}} = \mathrm{Gal}(\overline{k}_{\mathfrak{p}}/k_{\mathfrak{p}})$
tubular n.b.d of a link $V_{\mathcal{L}} = V_{\mathcal{K}_1} \sqcup \cdots \sqcup V_{\mathcal{K}_r}$ boundary tori $\partial V_{\mathcal{L}} = \partial V_{\mathcal{K}_1} \sqcup \cdots \sqcup \partial V_{\mathcal{K}_r}$	union of $\mathfrak{p}_i$-adic integer rings $V_S = \mathrm{Spec}(\mathcal{O}_{\mathfrak{p}_1}) \sqcup \cdots \sqcup \mathrm{Spec}(\mathcal{O}_{\mathfrak{p}_r})$ union of $\mathfrak{p}_i$-adic fields $\partial V_S = \mathrm{Spec}(k_{\mathfrak{p}_1}) \sqcup \cdots \sqcup \mathrm{Spec}(k_{\mathfrak{p}_r})$
link complement $X_{\mathcal{L}} = \overline{M} \setminus \mathrm{Int}(V_{\mathcal{L}})$ link group $\Pi_{\mathcal{L}} = \pi_1(X_{\mathcal{L}})$	complement of a finite set of primes $\overline{X}_S = \overline{X}_k \setminus S$ maximal Galois group with given ramification $\Pi_S = \mathrm{Gal}(k_S/k)$

Others

(1) 关于 Jones 多项式，与数学物理 (3 维规范场论，量子群，拓扑量子场论等) 紧密相关：Jones 多项式可源于 Hecke 代数迹，本质上为量子群表示的量子迹。
(2) 对 HOMFLY 多项式进行范畴化我们可以得到 Khovanov-Rozansky 同调，将其推广到四维流形里可以得到重要的不变量 Skein lasagna modules.
(3) 对已有的纽结不变量可以挖掘出新的不变量，比如对一个纽结的 Khovanov 同调里提取 number 得到 Rasmussen s-invariant, 其中有一个著名的不等式关系：$|s(K)| \leq 2g(K)$, g(K) 为 smooth slice genus.
(4) fibered knot (补空间是否可以纤维化：参考 Hopf 纤维, 其中三叶结，8 字结都是 fibered knot) 与 Legendrian knot(与切触结构，辛几何有关) 是比较热门的纽结，在研究纽结的过程中发展出许多手术 (Surgery or cut/paste)，比如 Fintushel-Stern knot surgery. 最重要的是，纽结与光滑 Poincare 猜想有关。

Classic papers and recent progress

Direct question

What properties of knots can homological invariants such as Khovanov homology and knot Floer homology actually detect, and how can we construct bridges between knot invariants (for instance, the spectral sequences utilized by Nathan Dowlin)?

Two fundamental theories of surgery

3-manifolds: Heegaard splitting. 4-manifolds: Kirby calculus

Casson handles

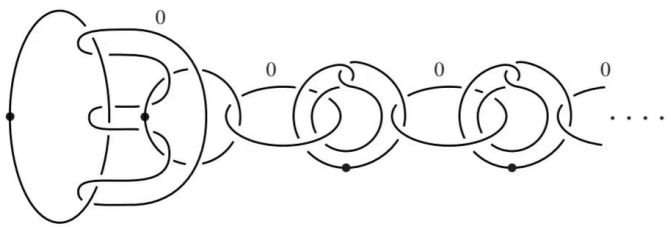


Figure 6.16. Exotic $\mathbb{R}^4$.

Classic papers and recent progress

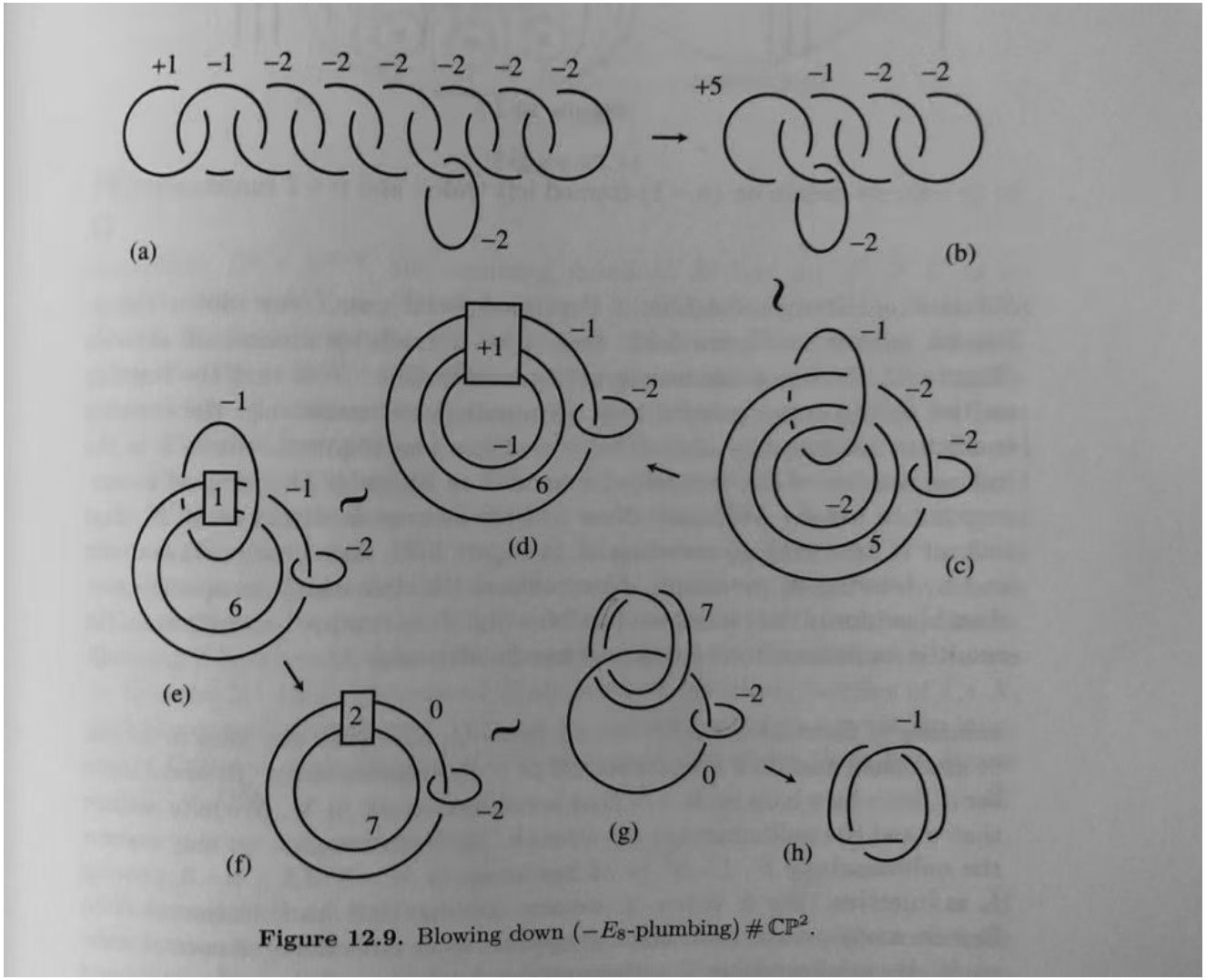


Figure 12.9. Blowing down ($-E_8$-plumbing) $\#\,\mathbb{CP}^2$.

图 29: Kirby calculus

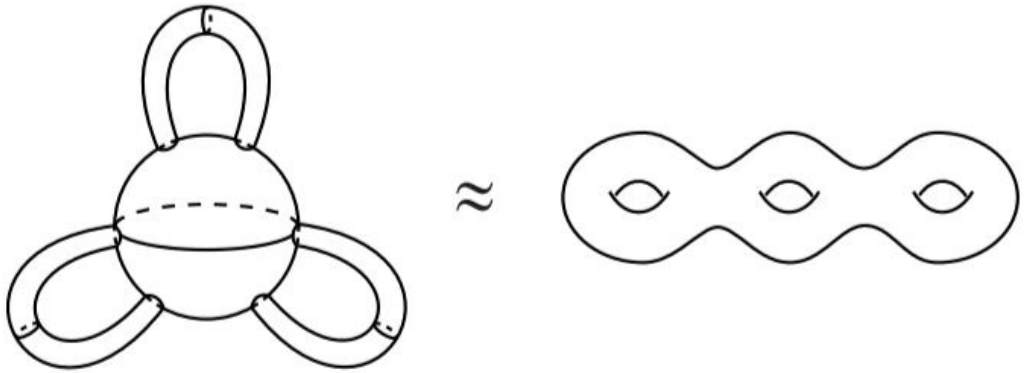


Figure 4.10. X_1 for a 3-manifold.

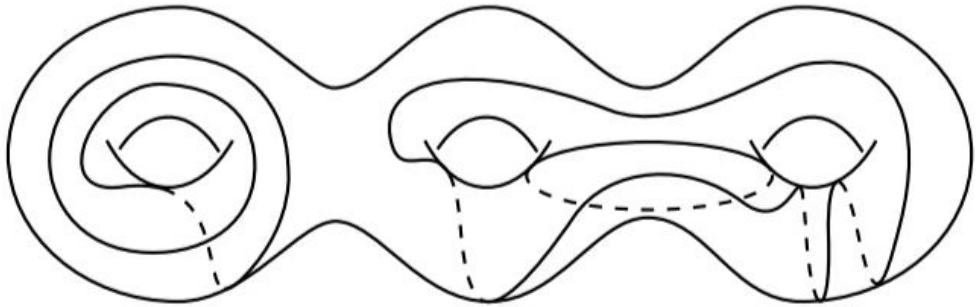


Figure 4.11. Heegaard diagram.

图 30: From Gompf: 《4-manifolds and kirby calculus》

Classic papers and recent progress

Problems on Low-dimensional Topology, 2015

Edited by T. Ohtsuki[1]

This is a list of open problems on low-dimensional topology with expositions of their history, background, significance, or importance. This list was made by editing manuscripts written by contributors of open problems to the problem session of the conference "Intelligence of Low-dimensional Topology" held at Research Institute for Mathematical Sciences, Kyoto University in May 20–22, 2015.

Contents

1 The slice-ribbon conjecture and Akbulut-Kirby's conjecture for 0-surgeries along knots **2**

2 Random links **3**

3 Braid ordering **4**

4 Relation between knots in S^3 and knots in $\mathbb{R}P^3$ **6**

5 Surgery presentations of genus two 3-manifolds **6**

6 Tight contact structures on 3-manifolds with open book decompositions **6**

7 Random 3-manifolds and orderability **7**

8 Stable complexity and stable presentation length for 3-manifolds **8**

9 Invariants of knots and 3-manifolds with representations of their fundamental groups **9**

The L-space Conjecture (BGW): Suppose Y is an irreducible rational homology 3-sphere. Then the following statements are equivalent:

(1) Y is a not an L-space (i.e. the Heegaard Floer homology of Y is not "simple"),

(2) $\pi_1(Y)$ is left-orderable, and

(3) Y admits a taut foliation.

Conclusion

“著名”辛几何学者 @ 李归农（烟花不堪剪）曾说：

Papepigs
编辑于 2025年3月20日 18:17

现在想来，2000年左右真是几何拓扑领域飞速发展的时代，Ozsváth-Szabó引进了Heegaard-Floer homology，Eliashberg-Givental-Hofer开创了symplectic field theory，Fukaya-Ono发明了Kuranishi structure的语言，还有Donaldson-Thomas理论等等。当时是定义和引入新的几何/拓扑不变量的一个高潮，许多文章并不解决问题，而是专注于引进新的不变量，而引进不变量的方式也都大同小异，就是要处理模空间，研究上面的相交理论，解决compactness和transversality等问题。

Papepigs
2025年6月19日 05:40

等我彻底离开世俗机构之后，打算把自己感兴趣的和解决不了的问题都写出来，也欢迎其他辛几何和辛拓扑领域的人分享他们感兴趣的问题，争取能搞一个属于辛拓扑的problem list。我目前感兴趣的问题，之前的动态里也讨论过一些，但是现在真正在做的东西主要有两样，一是关于rational homology plane的辛拓扑，二是关于代数簇中的cylinder，有兴趣了解的可以看一下这个survey：https://ems.press/journals/emss/articles/2504061。
我现在大致知道用什么工具去做能得到一些结果，但是不知道最终能做成什么样子，希望能尽可能做到完整的结果再和大家分享。这些东西都不是热门方向，甚至已经完全脱离我这个圈子的主流了，但是这些问题是我本人感兴趣很久的问题，如果再不去研究它们，以后可能也没机会了。

Conclusion

一切故事的开始，都起源于大二上学期那个平静的午后，偶然在数学 QQ 群文件里看到的一本书：姜伯驹《绳圈的数学》。

一入纽结深似海，knot theory 如一条庞杂无比的河流，比原来想象的还要复杂，一打开 arXiv 的 GT 区，60% 都是 knot theory, 现在也发展出来了许多更高阶，精细的代数不变量, 与 3 维 or 4 维流形也越来越紧密。**后来的人总是会再走前辈们走过的路，看到他们曾经经历过的风景。对于纽结，我看到了这条路的尽头是代数 (同调不变量, 表示论)，是物理 (TQFT)，是应用 (TDA：拓扑数据分析)，是生物 (DNA)!**

非常感谢大家的聆听，不足之处还望大家多多指教!